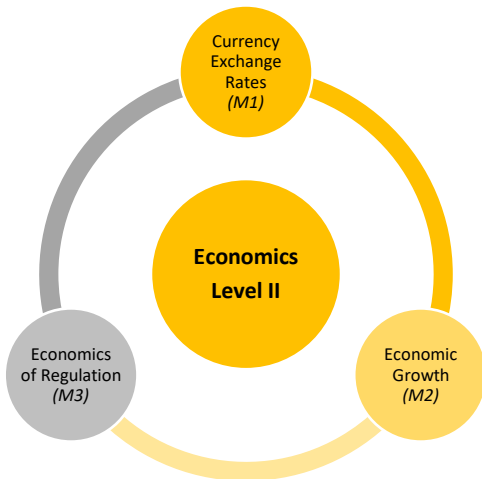
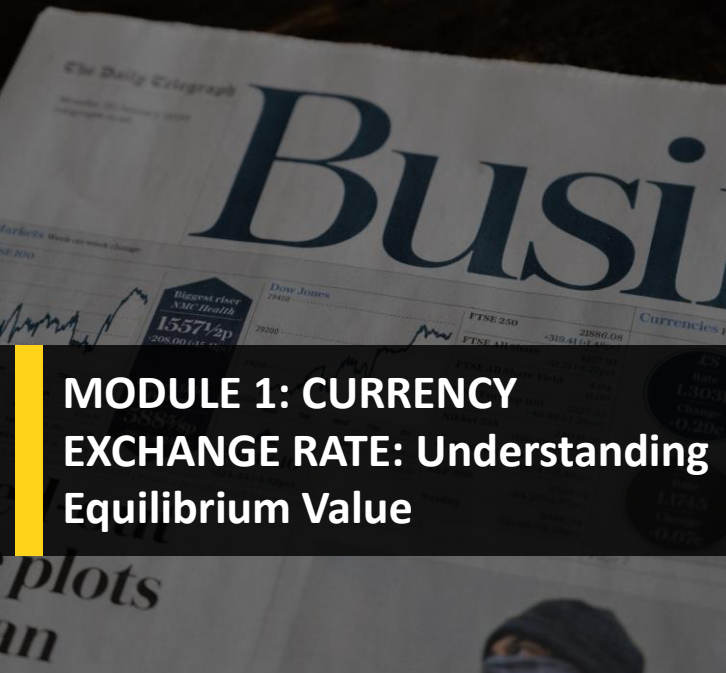


OVERVIEW





MODULE 1: CURRENCY EXCHANGE RATE: Understanding Equilibrium Value

LEARNING OUTCOMES

- 1.a.** Calculate and interpret the bid-offer spread on a spot or forward currency quotation and describe the factors that affect the bid-offer spread
- 1.b.** Identify a triangular arbitrage opportunity and calculate its profit, given the bid-offer quotations for three currencies
- 1.c.** Explain spot and forward rates and calculate the forward premium/discount for a given currency
- 1.d.** Calculate the mark-to-market value of a forward contract
- 1.e.** Explain international parity conditions (covered and uncovered interest rate parity, forward rate parity, purchasing power parity, and the international Fisher effect)
- 1.f.** Describe relations among the international parity conditions
- 1.g.** Evaluate the use of the current spot rate, the forward rate, purchasing power parity, and uncovered interest parity to forecast future spot exchange rates

LEARNING OUTCOMES

- 1.h.** Explain approaches to assessing the long-run value of an exchange rate
- 1.i.** Describe the carry trade and its relation to uncovered interest rate parity and calculate the profit from a carry trade
- 1.j.** Explain how flows in the balance of payment accounts affect currency exchange rates
- 1.k.** Explain the potential effects of monetary and fiscal policy on exchange rates
- 1.l.** Describe objectives of central bank or government intervention and capital controls and describe the effectiveness of intervention and capital controls
- 1.m.** Describe warning signs of a currency crisis

OVERVIEW

Currency exchange rate

1.a. Calculate and Interpret the spread

1.b. Identify arbitrage opportunity

1.c. Calculate forward premium /discount

1.d. Calculate mark-to-market value

Bid-offer spread

Triangular arbitrage

Spot and forward rates

Currency forward contract

1.a. Describe factors affect the spread

1.b. Calculate arbitrage profit

1.g. Evaluate the use to forecast future spot exchange rate

Covered IR parity

Uncovered IR parity

Forward rate parity

Purchasing power parity

International Fisher effect

1.e. Explain international parity conditions

1.f. Describe relations among the international parity conditions

1.h. Assessing the long-run fair value

1.i. Describe carry trade

Calculate the profit from a trade

1.j,k. Effect on exchange rate

1.j. Flows in the BOP accounts

1.k. Monetary and fiscal policy

1.l. Government intervention, capital controls

Objectives

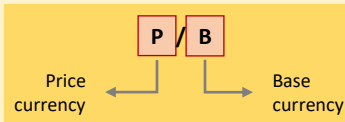
Effectiveness

1.m. Describe warning signs of a currency crisis

[LOS 1.a] Calculate and interpret the bid-offer spread on a spot or forward currency quotation and describe the factors that affect the bid-offer spread

1.

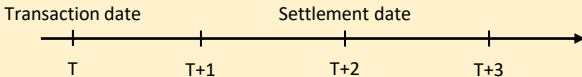
Exchange rate quotation convention



→ **Exchange rate:** Price of base currency (B) expressed in terms of the price currency (P).

E.g.: USD/EUR = 1.1650 → It costs 1.1650 US dollars to buy 1 euro

Spot exchange rate: The currency exchange rate for immediate delivery, the settlement takes place on the second business day after the trade (T+2).



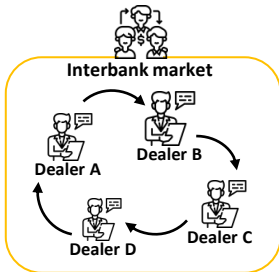
Note: The standard spot settlement exceptionally for CAD/USD is T+1.

[LOS 1.a] Calculate and interpret the bid-offer spread on a spot or forward currency quotation and describe the factors...

2.

Bid-offer spread

Interbank market vs. Retail market



Interbank market (or wholesale market): Dealers buy and sell foreign exchange among themselves.



Retail market: A dealer facilitate the trading of currencies for retail clients.

[LOS 1.a] Calculate and interpret the bid-offer spread on a spot or forward currency quotation and describe the factors...

2.

Bid-offer spread

Dealers offer their fellow dealers slightly more favorable rates.

→ The bid-offer spread is usually narrower in the interbank market than in the retail market.

Bid price: What the counterparty is willing to pay for one unit of the base currency

Offer price: What the counterparty is willing to sell one unit of the base currency at



Spread: difference between the offer and bid price

How does a dealer cover a position?

[LOS 1.a] Calculate and interpret the bid-offer spread on a spot or forward currency quotation and describe the factors...

2.

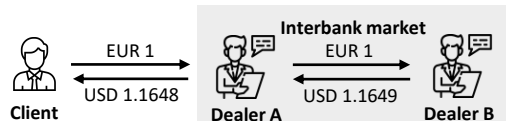
Bid-offer spread

How does a dealer cover a position?

Example:

Interbank: USD/EUR = 1.1649/1.1651

Dealer A: USD/EUR = 1.1648/1.1652



Profit for dealer A
 $= 1.1649 - 1.1648$
 $= 0.0001$ (1 pip)

[LOS 1.a] Calculate and interpret the bid-offer spread on a spot or forward currency quotation and describe the factors...

2.
Bid-offer spread
**Factors
affecting
bid-offer
spread**
**Liquidity of
the market**
**Currency
pair
involved**

Multiple bids and offers → High liquidity

**Time of
day**

Business hours overlap

**Market
volatility**

Lower uncertainty
→ Lower risk

Lower
spread

**Size of
transaction**

- Large transaction (> 50 million units) → Difficult for dealer to offset the position
- Retail transaction (< 1 million units)

Higher
spread

**Dealer –
client
relationship**

Favorable business relationship
Client's good credit profile

Lower
spread

[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

1.
Cross-rates
CROSS RATE

The exchange rate between two currencies implied by their exchange rates with a common third currency.


EXCHANGE RATE

The price of one currency in terms of another

Example 1: USD/EUR = 1.1649/1.1651; JPY/USD = 105.39/105.41
Calculate the implied JPY/EUR cross rate.

[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

1.

Cross-rates

Example 1: USD/EUR = 1.1649/1.1651; JPY/USD = 105.39/105.41
Calculate the implied JPY/EUR cross rate.

How to calculate cross rate?

1. Determine the price and base currency of the required cross-rate, written as a fraction

$$\text{JPY/EUR} \rightarrow \frac{\text{JPY}}{\text{EUR}}$$

2. Determine the “third currency”

The third currency is USD, given the existing exchange rates.

3. Follow the fraction rules: split the first fraction into two other fractions (exchange rates), so that the third currency can be crossed out

$$\frac{\text{JPY}}{\text{EUR}} = \frac{\text{JPY}}{\cancel{\text{USD}}} \times \frac{\cancel{\text{USD}}}{\text{EUR}}$$

4. Multiply the bids and offers of the two component fractions

$$\begin{aligned} \frac{\text{JPY}}{\text{EUR}_{\text{bid}}} &= \frac{\text{JPY}}{\text{USD}_{\text{bid}}} \times \frac{\text{USD}}{\text{EUR}_{\text{bid}}} = 105.39 \times 1.1649 = 122.77 \\ \frac{\text{JPY}}{\text{EUR}_{\text{offer}}} &= \frac{\text{JPY}}{\text{USD}_{\text{offer}}} \times \frac{\text{USD}}{\text{EUR}_{\text{offer}}} = 105.41 \times 1.1651 = 122.81 \\ &\rightarrow \text{JPY/EUR} = 122.77/122.81 \end{aligned}$$

[LOS 1.b] Identify a triangular arbitrage opportunity and calculate its profit, given the bid-offer quotations for three currencies

1.

Cross-rates

Example 2: USD/EUR = 1.1649/1.1651; USD/GBP = 1.2302/1.2304

Calculate the implied GBP/EUR cross rate.

How to calculate asymmetric cross rate?

1. Determine the price and base currency of the required cross-rate, written as a fraction

$$\text{GBP/EUR} \rightarrow \frac{\text{GBP}}{\text{EUR}}$$

2. Determine the third currency

The third currency is USD, given the existing exchange rates.

3. Follow the fraction rules: split the first fraction into two other fractions (exchange rates), so that the third currency can be crossed out. If a fraction is not given, invert as follows:

$$\frac{\text{GBP}}{\text{EUR}} = \frac{\text{GBP}}{\text{USD}} \times \frac{\text{USD}}{\text{EUR}}; \text{ GBP/USD is not given}$$

Invert USD/GBP quote → get GBP/USD quote:

$$\left(\frac{\text{GBP}}{\text{USD}}\right)_{\text{bid}} = \frac{1}{\left(\frac{\text{USD}}{\text{GBP}}\right)_{\text{offer}}} = \frac{1}{1.2304} = 0.8127$$

$$\left(\frac{\text{GBP}}{\text{USD}}\right)_{\text{offer}} = \frac{1}{\left(\frac{\text{USD}}{\text{GBP}}\right)_{\text{bid}}} = \frac{1}{1.2302} = 0.81288$$

[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

1.

Cross-rates

Example 2: USD/EUR = 1.1649/1.1651; USD/GBP = 1.2302/1.2304

Calculate the implied GBP/EUR cross rate.

How to calculate asymmetric cross rate?

4. *Multiply the bids and offers of the two component fractions*

$$\frac{\text{GBP}}{\text{EUR}_{\text{bid}}} = \frac{\text{GBP}}{\text{USD}_{\text{bid}}} \times \frac{\text{USD}}{\text{EUR}_{\text{bid}}} = 0.8127 \times 1.1649 = 0.9467$$

$$\frac{\text{GBP}}{\text{EUR}_{\text{offer}}} = \frac{\text{GBP}}{\text{USD}_{\text{offer}}} \times \frac{\text{USD}}{\text{EUR}_{\text{offer}}} = 0.81288 \times 1.1651 = 0.9471$$

Tips: Transform to get the largest bid-offer spread of the required cross-rate

[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

2.

Triangular arbitrage

Arbitrage principle

If identical financial products are priced differently, then market participants will buy the cheaper one and sell the more expensive one.

Locational arbitrage: the most common type of forex arbitrage and involve two currencies. (*example 1*)

Arbitrage constraints

01

Without these conditions of exchange rates

Dealer's bid < Interbank offer

Dealer's offer > Interbank bid

02

Dealer's cross-rate bids < Implied Interbank cross-rate offers

Dealer's cross-rate offers > Implied Interbank cross-rate bids

Without these conditions of cross-rates

Triangular arbitrage: the result of a discrepancy between three foreign currencies that occurs when the currency's exchange rates do not exactly match up. (*example 2*)

[LOS 1.b] Identify a triangular arbitrage opportunity and calculate its profit, given the bid-offer quotations for three currencies

2.

Triangular arbitrage

Example 1: Locational arbitrage

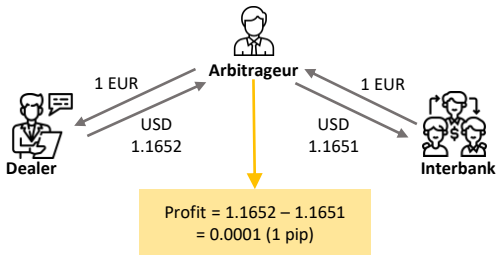
Interbank USD/EUR = 1.1649/1.1651

Dealer USD/EUR = 1.1652/1.1654

Interbank offer < Dealer's bid → *Violate constraint 1*

Strategy:

- Buy EUR in the Interbank market at USD 1.1651
- Sell EUR to dealer at USD 1.1652



[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

2. Triangular arbitrage

Example 2: Triangular arbitrage

Interbank market quotes:

- USD/AUD = 0.6000/0.6015
- USD/MXN = 0.0933/0.0935

Dealer's quote: MXN/AUD = 6.3000/6.3025

Is there an arbitrage opportunity? Compute the gains and losses of possible trading strategies, starting with USD 1 million.

$$\frac{\text{MXN}}{\text{AUD}} = \frac{\text{MXN}}{\text{USD}} \times \frac{\text{USD}}{\text{AUD}}$$

Need to invert from USD/MXN

Step 1: Compute implied cross rate (asymmetric cross rate)

MXN/AUD_{bid}

$$\begin{aligned} \frac{\text{MXN}}{\text{USD}_{\text{bid}}} &= \frac{1}{\frac{\text{USD}}{\text{MXN}_{\text{offer}}}} = \frac{1}{0.0935} \\ &= 10.7 \end{aligned}$$

The implied cross bid rate is:

$$\begin{aligned} \frac{\text{MXN}}{\text{AUD}_{\text{bid}}} &= \frac{\text{USD}}{\text{AUD}_{\text{bid}}} \times \frac{\text{MXN}}{\text{USD}_{\text{bid}}} \\ &= 0.6 \times 10.7 \\ &= 6.42 \end{aligned}$$

MXN/AUD_{offer}

$$\begin{aligned} \frac{\text{MXN}}{\text{USD}_{\text{offer}}} &= \frac{1}{\frac{\text{USD}}{\text{MXN}_{\text{bid}}}} = \frac{1}{0.0933} \\ &= 10.72 \end{aligned}$$

The implied cross ask rate is:

$$\begin{aligned} \frac{\text{MXN}}{\text{AUD}_{\text{offer}}} &= \frac{\text{USD}}{\text{AUD}_{\text{offer}}} \times \frac{\text{MXN}}{\text{USD}_{\text{offer}}} \\ &= 0.6015 \times 10.72 \\ &= 6.4481 \end{aligned}$$

→ MXN/AUD = 6.4200/6.4481

[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

2.

Triangular arbitrage

Example 2: Triangular arbitrage

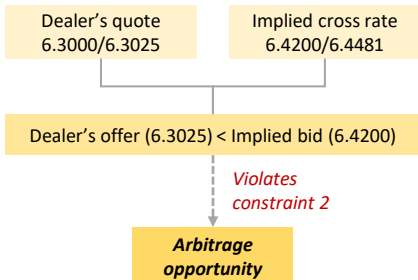
Interbank market quotes:

- USD/AUD = 0.6000/0.6015
- USD/MXN = 0.0933/0.0935

Dealer's quote: MXN/AUD = 6.3000/6.3025

Is there an arbitrage opportunity? Compute the gains and losses of possible trading strategies, starting with USD 1 million.

Step 2: Compare the implied MXN/AUD cross rate with the dealer's quote:



[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

2.

Triangular arbitrage

Example 2: Triangular arbitrage

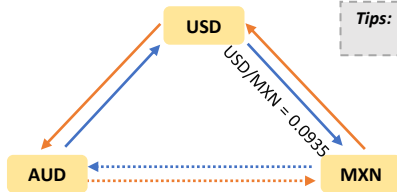
Interbank market quotes:

- USD/AUD = 0.6000/0.6015
- USD/MXN = 0.0933/0.0935

Dealer's quote: MXN/AUD = 6.3000/6.3025

Is there an arbitrage opportunity? Compute the gains and losses of possible trading strategies, starting with USD 1 million.

Step 3: Compute the arbitrage profit:



Tips: Up the bid and multiply
Down the ask (offer) and divide

Clockwise: (1) USD → MXN

Down the ask and divide

Start with
USD

Sell USD
Buy MXN

Down $\frac{\text{USD}}{\text{MXN}}$

Use the
offer rate
(0.0935)

Divide

$10,000,000 / 0.0935 = 10,695,187 \text{ MXN}$

[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

2.

Triangular arbitrage

Example 2: Triangular arbitrage

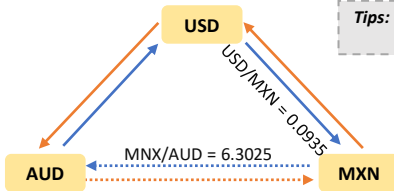
Interbank market quotes:

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Dealer's quote: MXN/AUD = 6.3000/6.3025

Is there an arbitrage opportunity? Compute the gains and losses of possible trading strategies, starting with USD 1 million.

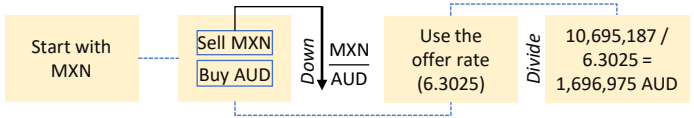
Step 3: Compute the arbitrage profit:



Tips: Up the bid and multiply
Down the ask (offer) and divide

Clockwise: (2) MXN → AUD

Down the ask and divide



[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

2.

Triangular arbitrage

Example 2: Triangular arbitrage

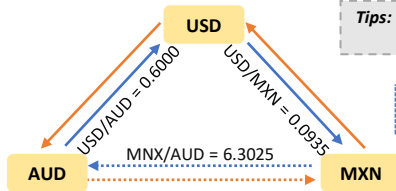
Interbank market quotes:

- USD/AUD = 0.6000/0.6015
- USD/MXN = 0.0933/0.0935

Dealer's quote: MXN/AUD = 6.3000/6.3025

Is there an arbitrage opportunity? Compute the gains and losses of possible trading strategies, starting with USD 1 million.

Step 3: Compute the arbitrage profit:



Tips: Up the bid and multiply
Down the ask (offer) and divide

$$\text{Profit} = 1,018,185 - 1,000,000 = 18,185 \text{ USD}$$

Clockwise: (3) AUD → USD

Up the bid and multiply

Start with
AUD

Sell AUD
Buy USD

Up
 $\frac{\text{USD}}{\text{AUD}}$

Use the bid
rate
(0.6000)

Multiply

$$1,696,975 \times 0.6000 = 1,018,185 \text{ USD}$$

[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

2. Triangular arbitrage

Example 2: Triangular arbitrage

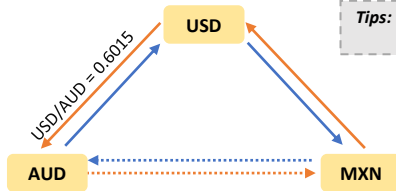
Interbank market quotes:

- USD/AUD = 0.6000/0.6015
- USD/MXN = 0.0933/0.0935

Dealer's quote: MXN/AUD = 6.3000/6.3025

Is there an arbitrage opportunity? Compute the gains and losses of possible trading strategies, starting with USD 1 million.

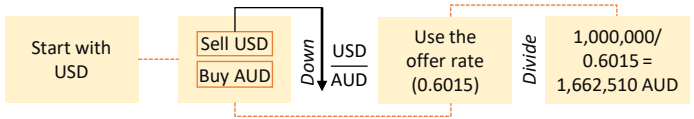
Step 3: Compute the arbitrage profit:



Tips: Up the bid and multiply
Down the ask (offer) and divide

Counter-clockwise: (1) USD → AUD

Down the ask and divide



[LOS 1.b] Identify a triangular arbitrage opportunity and calculate its profit, given the bid-offer quotations for three currencies

2.

Triangular arbitrage

Example 2: Triangular arbitrage

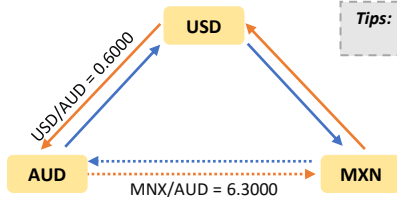
Interbank market quotes:

- USD/AUD = 0.6000/0.6015
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Dealer's quote: MXN/AUD = 6.3000/6.3025

Is there an arbitrage opportunity? Compute the gains and losses of possible trading strategies, starting with USD 1 million.

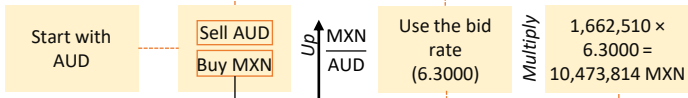
Step 3: Compute the arbitrage profit:



Tips: Up the bid and multiply
Down the ask (offer) and divide

Counter-clockwise: (2) AUD → MXN

Up the bid and multiply



[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

2.

Triangular arbitrage

Example 2: Triangular arbitrage

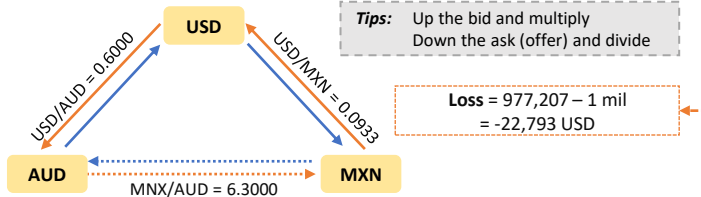
Interbank market quotes:

- USD/AUD = 0.6000/0.6015
- USD/MXN = 0.0933/0.0935

Dealer's quote: MXN/AUD = 6.3000/6.3025

Is there an arbitrage opportunity? Compute the gains and losses of possible trading strategies, starting with USD 1 million.

Step 3: Compute the arbitrage profit:



Counter-clockwise: (3) MXN → USD

Up the bid and multiply



[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

2.

Triangular arbitrage

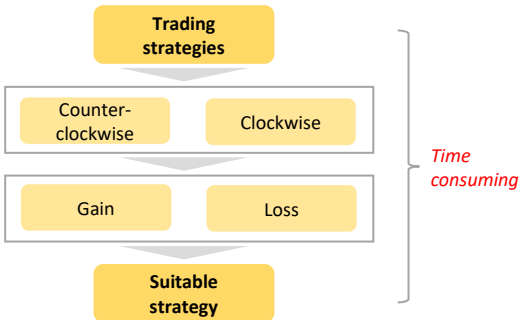
Example 2: Triangular arbitrage

Interbank market quotes:

- USD/AUD = 0.6000/0.6015
- USD/MXN = 0.0933/0.0935

Dealer's quote: MXN/AUD = 6.3000/6.3025

Is there an arbitrage opportunity? Compute the gains and losses of possible trading strategies, starting with USD 1 million.



How to find arbitrage profit without computing both strategies?

[LOS 1.b] Identify a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

2.

Triangular arbitrage

How to find arbitrage profit without computing both strategies?

Example 2: Triangular arbitrage

Interbank market quotes:

- USD/AUD = 0.6000/0.6015
- USD/MXN = 0.0933/0.0935

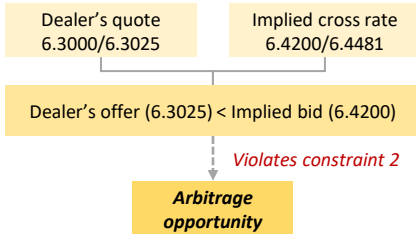
Dealer's quote: MXN/AUD = 6.3000/6.3025

Is there an arbitrage opportunity? Compute the gains and losses of possible trading strategies, starting with USD 1 million.

Step 1: Compute implied cross rates (results from previous calculation)

→ MXN/AUD = 6.4200/6.4481

Step 2: Compare the implied cross rates with the dealer's quote



[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

2. Triangular arbitrage

How to find arbitrage profit without computing both strategies?

Example 2: Triangular arbitrage

Interbank market quotes:

- USD/AUD = 0.6000/0.6015
- USD/MXN = 0.0933/0.0935

Dealer's quote: MXN/AUD = 6.3000/6.3025

Is there an arbitrage opportunity? Compute the gains and losses of possible trading strategies, starting with USD 1 million.

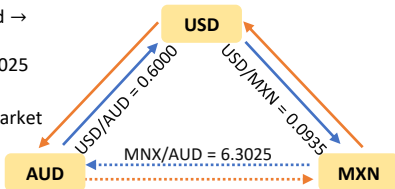
Step 3: Determine appropriate strategy from the quotes



Because dealer's offer < implied bid → arbitrageur will exploit this by:

- Buying AUD from dealer at 6.3025 (MXN → AUD)
- Selling AUD in the interbank market

→ Going for **Clockwise strategy**.



[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

2.

Triangular arbitrage

How to find arbitrage profit without computing both strategies?

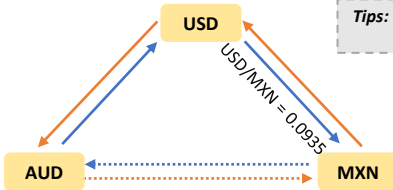
Example 2: Triangular arbitrage

Interbank market quotes:

- USD/AUD = 0.6000/0.6015
- USD/MXN = 0.0933/0.0935

Dealer's quote: MXN/AUD = 6.3000/6.3025

Step 4: Compute the arbitrage profit:



Tips: Up the bid and multiply
Down the ask (offer) and divide

Clockwise: (1) USD → MXN

Down the ask and divide

Start with
USD

Sell USD
Buy MXN

Down
 $\frac{\text{USD}}{\text{MXN}}$

Use the
offer rate
(0.0935)

Divide

$10,000,000 / 0.0935 = 10,695,187 \text{ MXN}$

[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

2. Triangular arbitrage

How to find arbitrage profit without computing both strategies?

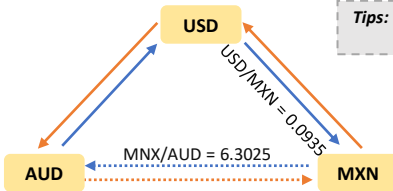
Example 2: Triangular arbitrage

Interbank market quotes:

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- USD/MXN = 0.0933/0.0935

Dealer's quote: MXN/AUD = 6.3000/6.3025

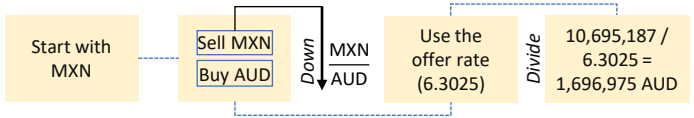
Step 4: Compute the arbitrage profit:



Tips: Up the bid and multiply
Down the ask (offer) and divide

Clockwise: (2) MXN → AUD

Down the ask and divide



[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

2.

Triangular arbitrage

How to find arbitrage profit without computing both strategies?

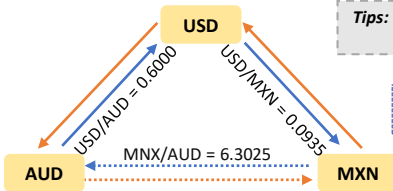
Example 2: Triangular arbitrage

Interbank market quotes:

- USD/AUD = 0.6000/0.6015
- USD/MXN = 0.0933/0.0935

Dealer's quote: MXN/AUD = 6.3000/6.3025

Step 4: Compute the arbitrage profit:



Tips: Up the bid and multiply
Down the ask (offer) and divide

$$\text{Profit} = 1,018,185 - 1,000,000 = 18,185 \text{ USD}$$

✓ **Clockwise strategy does provide arbitrage profit.**

Clockwise: (3) AUD → USD

Up the bid and multiply

Start with
AUD

Sell AUD
Buy USD

Up
 $\frac{\text{USD}}{\text{AUD}}$

Use the bid
rate
(0.6000)

Multiply

$$1,696,975 \times 0.6000 = 1,018,185 \text{ USD}$$

[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

2. Triangular arbitrage

Summary: Calculate triangular arbitrage profit

1. Compute the implied cross-rates (refer to slide 11 to 13)

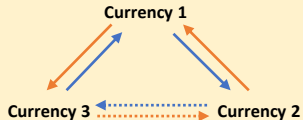
- Determine the price and base currency of the cross-rate
- Split into 2 component fractions following the rules of fraction
- Multiply the bids and offers to get the cross-rate quotes

2. Compare implied cross-rate with dealer's quote

Compare to see whether **Constraint 2** is violated (refer to slide 14)

3. Determine the appropriate strategy

- Draw a triangle of currencies with dealer's quote at the bottom
- Using the violated quotes to determine the direction of trade under the dealer's quotes
 - *Dealer's offer < Implied bid*
→ Buy from dealer, sell in interbank market
 - *Implied offer < Dealer's bid*
→ Buy from interbank market, sell to dealer
- Determine overall trading strategy (**clockwise** or **counter-clockwise**)



4. Calculate the arbitrage profit

- Start with the initial investment currency
- Follow the tips: *Up the bid and multiply*
Down the ask and divide
- Find the cash inflow after converting back to initial currency and compute profit

[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

Knowledge check
Example: Bid-offer rates

Spot rate quotes in the interbank market:

USD/EUR	1.1649/1.1651
JPY/USD	105.39/105.41
CAD/USD	1.3199/1.3201
SEK/USD	9.6300/9.6302

Recall: Calculating asymmetric cross rate:

1. Determine price and base currency

2. Determine the third currency

3. Follow the fraction rules

4. Multiply the bids and offers of the two component fractions

1. What is the bid-offer on the JPY/CAD cross rate implied by the interbank market?

- A. 78.13/78.17
- B. 79.85/79.85
- C. 79.84/79.86

[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

Knowledge check

Example: Bid-offer rates

Spot rate quotes in the interbank market:

USD/EUR 1.1649/1.1651

JPY/USD 105.39/105.41

CAD/USD 1.3199/1.3201

SEK/USD 9.6300/9.6302

Recall: Calculating asymmetric cross rate:

1. What is the bid-offer on the JPY/CAD cross rate implied by the interbank market?

A. 78.13/78.17

B. 79.85/79.85

C. 79.84/79.86

1. Determine price and base currency

$$\text{JPY/CAD} = \frac{\text{JPY}}{\text{CAD}}$$

2. Determine the third currency:

The third currency is USD

3. Follow the fraction rules

$$\frac{\text{JPY}}{\text{CAD}} = \frac{\text{JPY}}{\text{USD}} \times \frac{\text{USD}}{\text{CAD}}$$

$$\frac{\text{USD}}{\text{CAD}_{\text{bid}}} = \frac{1}{\frac{\text{CAD}}{\text{USD}_{\text{offer}}}} = \frac{1}{1.3201} = 0.75752$$

$$\frac{\text{USD}}{\text{CAD}_{\text{offer}}} = \frac{1}{\frac{\text{CAD}}{\text{USD}_{\text{bid}}}} = \frac{1}{1.3199} = 0.75763$$

4. Multiply the bids and offers of the two component fractions

$$\frac{\text{JPY}}{\text{CAD}_{\text{bid}}} = \frac{\text{JPY}}{\text{USD}_{\text{bid}}} \times \frac{\text{USD}}{\text{CAD}_{\text{bid}}} = 105.39 \times 0.75752 = 79.84$$

$$\frac{\text{JPY}}{\text{CAD}_{\text{offer}}} = \frac{\text{JPY}}{\text{USD}_{\text{offer}}} \times \frac{\text{USD}}{\text{CAD}_{\text{offer}}} = 105.41 \times 0.75763 = 79.86$$

→ JPY/CAD = 79.84/79.86

[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

Knowledge check

Example: Bid-offer rates

Spot rate quotes in the interbank market:

USD/EUR	1.1649/1.1651
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CAD/USD	1.3199/1.3201
SEK/USD	9.6300/9.6302

Step 1: Compute the implied cross-rates

Step 2: Compare with dealer's quote

Step 3: Determine the appropriate strategy

2. If a dealer quoted a bid-offer rate of 79.81/79.83 in JPY/CAD, then a triangular-arbitrage would involve buying:

- CAD in the interbank market and selling it to the dealer, for a profit of JPY 0.01 per CAD.
- JPY from the dealer and selling it in the interbank market, for a profit of CAD 0.01 per JPY.
- CAD from the dealer and selling it in the interbank market, for a profit of JPY 0.01 per CAD.

Step 4: Calculate the arbitrage profit

[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

Knowledge check

Example: Bid-offer rates

Spot rate quotes in the interbank market:

USD/EUR	1.1649/1.1651
JPY/USD	105.39/105.41
CAD/USD	1.3199/1.3201
SEK/USD	9.6300/9.6302

Step 1: Compute the implied cross-rates

From the previous question, the implied cross rate for JPY/CAD is 79.84/79.86.

Step 2: Compare with dealer's quote

Dealer quote
79.81/79.83

Implied cross rate
79.84/79.86

2. If a dealer quoted a bid-offer rate of 79.81/79.83 in JPY/CAD, then a triangular-arbitrage would involve buying:

- CAD in the interbank market and selling it to the dealer, for a profit of JPY 0.01 per CAD.
- JPY from the dealer and selling it in the interbank market, for a profit of CAD 0.01 per JPY.
- CAD from the dealer and selling it in the interbank market, for a profit of JPY 0.01 per CAD.

Dealer's offer (79.83) < Implied bid (79.86)

Violates constraint 2

**Arbitrage
opportunity**

[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

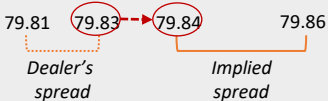
Knowledge check

Example: Bid-offer rates

Spot rate quotes in the interbank market:

USD/EUR	1.1649/1.1651
JPY/USD	105.39/105.41
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Step 3: Determine the appropriate strategy



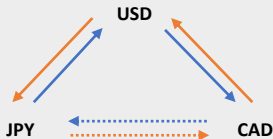
Because dealer's offer < implied bid
→ arbitrageur will exploit this by:

- Buying CAD from the dealer at 79.83 (JPY → CAD)
- Selling CAD in the interbank market

→ Going for Counter-clockwise strategy.

2. If a dealer quoted a bid-offer rate of 79.81/79.83 in JPY/CAD, then a triangular-arbitrage would involve buying:

- CAD in the interbank market and selling it to the dealer, for a profit of JPY 0.01 per CAD.
- JPY from the dealer and selling it in the interbank market, for a profit of CAD 0.01 per JPY.
- CAD from the dealer and selling it in the interbank market, for a profit of JPY 0.01 per CAD.



[LOS 1.b] *Identify* a triangular arbitrage opportunity and *calculate* its profit, given the bid-offer quotations for three currencies

Knowledge check

Example: Bid-offer rates

Spot rate quotes in the interbank market:

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JPY/USD	105.39/105.41
CAD/USD	1.3199/1.3201
SEK/USD	9.6300/9.6302

Step 4: Calculate the arbitrage profit



Because the question is asking for arbitrage profit either in JPY or CAD, we only need to use JPY/CAD cross-rates.

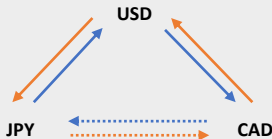
According to the strategy

- Buying CAD from the dealer at 79.83 (JPY → CAD)
- Selling CAD in the interbank market at 79.84 from the implied rate (CAD → USD → JPY)

→ Profit = 79.84 – 79.83 = JPY 0.01

2. If a dealer quoted a bid-offer rate of 79.81/79.83 in JPY/CAD, then a triangular-arbitrage would involve buying:

- CAD in the interbank market and selling it to the dealer, for a profit of JPY 0.01 per CAD.
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- CAD from the dealer and selling it in the interbank market, for a profit of JPY 0.01 per CAD.



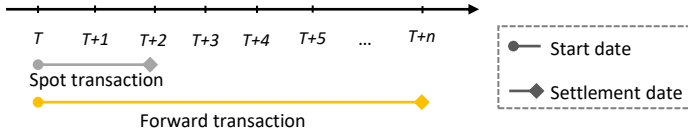
[LOS 1.c] *Explain* spot and forward rates and *calculate* the forward premium/discount for a given currency

1.

Forward exchange rates

Recall:

Spot exchange rate is the currency exchange rate for immediate delivery, the settlement takes place on the second business day after the trade ($T+2$).



Forward contracts

Agreements to exchange one currency for another on a future date at an **exchange rate agreed upon today**

Forward exchange rates

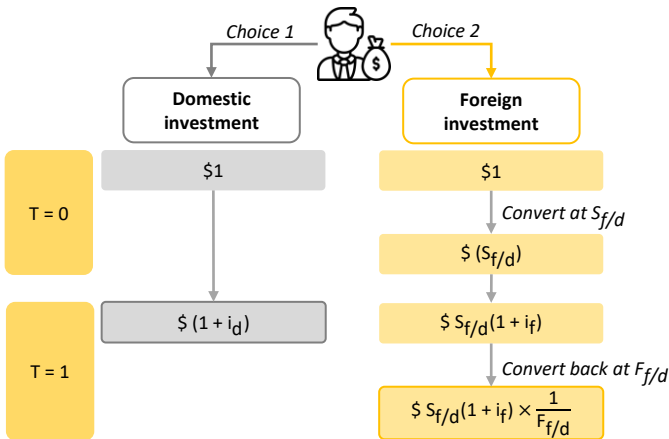
How to make profit from forward exchange rates?

[LOS 1.c] Explain spot and forward rates and *calculate* the forward premium/discount for a given currency

1. Forward exchange rates

How to make profit from forward exchange rates? - Covered interest rate parity

Investor is assumed to transact at a forward rate $F_{f/d}^*$ that was **locked in** at initiation – **covered**.



*For convenience, the notation for exchange rate will be foreign/domestic currency (f/d).

[LOS 1.c] Explain spot and forward rates and *calculate* the forward premium/discount for a given currency

1.

Forward exchange rates

How to make profit from forward exchange rates? - Covered interest rate parity

**Domestic
investment**

$$$(1 + i_d)$$

**Foreign
investment**

$$\$ S_{f/d}(1 + i_f) \times \frac{1}{F_{f/d}}$$



Arbitrage opportunity

Borrow internationally, invest domestically



**For convenience, the notation for exchange rate will be foreign/domestic currency (f/d).*

[LOS 1.c] *Explain* spot and forward rates and *calculate* the forward premium/discount for a given currency

1.

Forward exchange rates

How to make profit from forward exchange rates? - Covered interest rate parity

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Arbitrage opportunity

Borrow domestically, invest internationally



**For convenience, the notation for exchange rate will be foreign/domestic currency (f/d).*

[LOS 1.c] *Explain* spot and forward rates and *calculate* the forward premium/discount for a given currency

1.

Forward exchange rates

How to make profit from forward exchange rates? - Covered interest rate parity

Domestic
investment

$$\$(1 + i_d)$$

Foreign
investment

$$\$ S_{f/d}(1 + i_f) \times \frac{1}{F_{f/d}}$$

=

No riskless arbitrage profit



**Covered interest
rate parity**

$$F_{f/d} = S_{f/d} \times \frac{1 + i_f}{1 + i_d}$$

Covered: bound by arbitrage

*If it is possible to earn more interest in the foreign market than in the domestic market, then the forward discount for the foreign currency will **offset** the higher foreign interest rate*

[LOS 1.c] Explain spot and forward rates and *calculate* the forward premium/discount for a given currency

2.

Forward premium/discount

Covered interest rate parity

$$\begin{aligned}
 F_{A/B} &= S_{A/B} \times \frac{1 + i_A}{1 + i_B} \\
 F_{A/B} - S_{A/B} &= S_{A/B} \times \left(\frac{1 + i_A}{1 + i_B} - 1 \right) \\
 &= S_{A/B} \times \frac{1 + i_A - 1 - i_B}{1 + i_B} \\
 &= S_{A/B} \times \frac{i_A - i_B}{1 + i_B} \\
 &\text{with } (1 + i_B \approx 1)
 \end{aligned}$$

$$\begin{aligned}
 &\rightarrow F_{A/B} - S_{A/B} = S_{A/B} \times (i_A - i_B) \\
 &\rightarrow \underbrace{\frac{F_{A/B} - S_{A/B}}{S_{A/B}}}_{\text{Forward premium/discount}} = \underbrace{i_A - i_B}_{\text{Interest rate differential}}
 \end{aligned}$$

Forward
premium/
discount

Covered interest rate parity

Interest
rate
differential

Note: Covered interest rate parity is **no-arbitrage** condition.
Disparity will be quickly arbitrated away $\rightarrow$ CIRP always holds

$$\frac{F_{A/B} - S_{A/B}}{S_{A/B}} = i_A - i_B$$

$i_A > i_B \rightarrow$ B trades at forward premium ($F_{A/B} > S_{A/B}$)

$i_A < i_B \rightarrow$ B trades at forward discount ($F_{A/B} < S_{A/B}$)

[LOS 1.d] Calculate the mark-to-market value of a forward contract

1.

Forward contract



Forward contracts

Agreements to exchange one currency for another on a future date at an exchange rate agreed upon today.

Mark-to-market value reflects the *profit (or loss)* that would be realized from *closing out* the position at current market prices.

Example:

A market participant enter a forward contract: long GBP 10 million against AUD in six months at an “all-in” forward rate of 1.8100 AUD/GBP (*All-in forward rate is the sum of the spot rate and the scaled forward points*).

Three months later, the market participant wants to close out for this forward contract.

Assuming the bid-offer quotes for spot and forward points three months prior to the settlement date are:

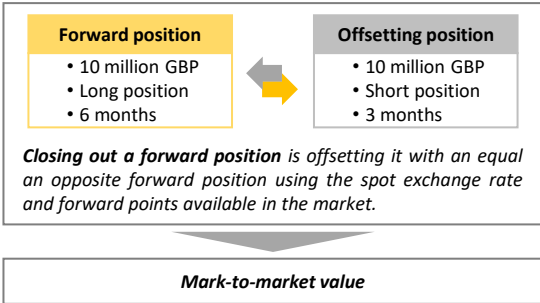
Spot rate (AUD/GBP)	1.8210/1.8215
Three-month points	130/140
Three-month AUD MRR	2.40%

Calculate the mark-to-market value of this contract in AUD (the price currency).

The process to calculate mark-to-market value is illustrated on the next slides.

[LOS 1.d] Calculate the mark-to-market value of a forward contract

2. Calculate mark-to-market value

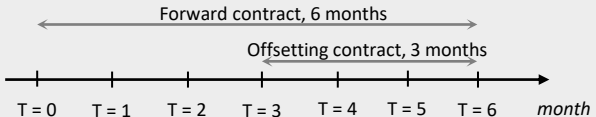


Illustration

Step 1: Create an offsetting forward position equal to the original forward position

Forward contract: long GBP 10 million against AUD

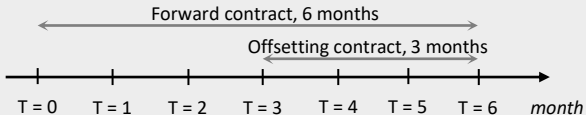
Closing out: Offset by short GBP 10 million, long AUD



[LOS 1.d] Calculate the mark-to-market value of a forward contract

2. Calculate mark-to-market value

Illustration (cont.)



Step 2: Determine appropriate all-in forward rate used for offsetting position

Spot rate AUD/GBP = 1.8210/1.8215 }
 3-month points: 130/140 } → Forward rate = 1.8340/1.8355

Step 3: Calculate cash flow at settlement day

At T = 0, enter forward contract at all-in rate
 AUD/GBP = 1.8100
 In 6 months, short 10 mil × 1.81 = AUD 18.1 mil

At T = 3, enter offsetting contract at forward bid rate
 AUD/GBP = 1.8340
 In 3 months, long 10 mil × 1.8340 = AUD 18.340 mil

Cash flow at T = 6:
 Net gain/loss = 18.340
 – 18.1
 = AUD 0.240 mil
 = AUD 240,000

Step 4: Calculate the present value of cash flow at settlement

The market participant want to close out the contract at T = 3 → Must discount to T = 3:

$$PV_3 = \frac{\text{AUD } 240,000}{1 + 2.4\% \times \frac{90}{360}} = \text{AUD } 238,569 \rightarrow \text{Mark-to-market value}$$

[LOS 1.d] Calculate the mark-to-market value of a forward contract

2. Calculate mark-to-market value

How to calculate Mark-to-market value at time t ?

1. Create an offsetting forward position equal to the original forward position

Initial forward contract	Offsetting forward contract
Long position Short position	Short position Long position

2. Determine appropriate all-in forward rate used for offsetting position

Spot rate at t	+	Forward premium/ discount	=	Forward rate at t
---------------------	---	---------------------------------	---	------------------------

Depending on offsetting position, determine whether to use bid or offer rate.

3. Calculate cash flow at settlement day

$$\text{Net cash flow} = (\text{Forward rate}_t - \text{Forward rate}_0) \times \text{Contract size}$$

(for the case of long forward original and short forward offset)

4. Calculate the present value of cash flow at settlement

$$\text{MTM value at } t = V_t = \frac{\text{Net cash flow}}{1 + R \times \left(\frac{\text{Days}}{360}\right)}$$

R is the discount rate of the currency of the cash flow
(usually the price currency).

For quick calculation, we can use the formula:

$$V_t = \frac{(\text{FP}_t - \text{FP}_0)(\text{Contract size})}{1 + R \left(\frac{\text{Days}}{360}\right)}$$

Where:

FP_t = Forward price at time t to sell the base

FP_0 = Forward price at time 0 to buy the base

[LOS 1.d] *Calculate* the mark-to-market value of a forward contract

2.
Calculate mark-to-market value

Knowledge check

Example: Mark-to-market value

Some time ago, Laurier Bay Capital, an investment fund based in LOS Angeles, hedged a long exposure to the New Zealand dollar by selling NZD 10 million forward against the USD in 6 months; the all-in forward price was 0.7900 (USD/ NZD). Three months prior to the settlement date, Laurier Bay wants to mark this forward position to market. The bid–offer for the USD/NZD spot rate, the three-month forward points, and the three-month MRRs (annualized) are as follows:

Spot rate (USD/NZD)	0.7825/0.7830
Three-month points	-12.1/-10.0
Three-month MRR (NZD)	3.31%
Three-month MRR (USD)	0.31%

The mark-to-market value for Laurier Bay's forward position is *closest to*:

- A. –USD 87,100
- B. +USD 77,437
- C. +USD 79,938

[LOS 1.d] Calculate the mark-to-market value of a forward contract

2. Calculate mark-to-market value

Knowledge check

Answer: C

1. Create an offsetting forward position

Initial forward contract	Offsetting forward contract
NZD 10 million Short NZD 6 months	NZD 10 million Long NZD 3 months

2. Determine appropriate all-in forward rate for offsetting position

Offset: Short USD, long NZD → Use offer rate (0.7830)

All-in forward rate = $0.7830 - 0.0010 = 0.7820$

3. Calculate cash flow at settlement day

Note: On settlement day, Laurier Bay receive USD from initial forward contract and pay USD in the offsetting contract

Net cash flow = $(- \text{Forward rate}_t + \text{Forward rate}_0) \times \text{Contract size}$
 $= (-0.7820 + 0.7900) \times \text{NZD } 10 \text{ million} = \text{USD } 80,000$

4. Calculate the present value of cash flow at settlement

Currency of the cash flow is USD → $R_{\text{USD}} = 0.31\%$

$$V_t = \frac{\text{Net cash flow}}{1 + R_{\text{USD}} \times \frac{\text{Days}}{360}} = \frac{80,000}{1 + 0.31\% \times \frac{90}{360}} = 79,938 \text{ USD}$$

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

- 1.e.** Explain international parity conditions (covered and uncovered interest rate parity, forward rate parity, purchasing power parity, and the international Fisher effect)
- 1.f.** Describe relations among the international parity conditions
- 1.g.** Evaluate the use of the current spot rate, the forward rate, purchasing power parity, and uncovered interest parity to forecast future spot exchange rates
- 1.h.** Explain approaches to assessing the long-run fair value of an exchange rate

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

International parity conditions

1

Covered interest rate parity

2

Uncovered interest rate parity

3

Forward rate parity

4

Purchasing power parity

5

The International Fisher effect

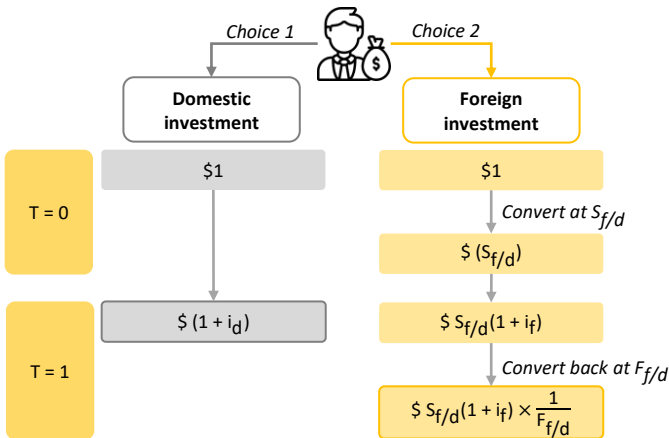
The Fisher effect, Real Interest Rate parity, and International Parity Conditions

1.

Covered and Uncovered interest rate parity

Covered interest rate parity (*as demonstrated in LOS 1.c*)

Investor is assumed to transact at a forward rate $F_{f/d}^*$ that was **locked in** at initiation – **covered**.



*For convenience, the notation for exchange rate will be foreign/domestic currency (f/d).

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

1.

Covered and Uncovered interest rate parity

Covered interest rate parity (*as demonstrated in LOS 1.c*)

Domestic investment

$$$(1 + i_d)$$

Foreign investment

$$\$ S_{f/d}(1 + i_f) \times \frac{1}{F_{f/d}}$$



Arbitrage opportunity

Borrow internationally, invest domestically



**For convenience, the notation for exchange rate will be foreign/domestic currency (f/d).*

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

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Covered and Uncovered interest rate parity

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Foreign investment

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The Fisher effect, Real Interest Rate parity, and International Parity Conditions

1.

Covered and Uncovered interest rate parity

Covered interest rate parity (as demonstrated in LOS 1.c)

Domestic
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$$$(1 + i_d)$$

Foreign
investment

$$\$ S_{f/d}(1 + i_f) \times \frac{1}{F_{f/d}}$$

=

No riskless arbitrage profit



**Covered interest
rate parity**

$$F_{f/d} = S_{f/d} \times \frac{1 + i_f}{1 + i_d}$$

Covered: bound by arbitrage

*If it is possible to earn more interest in the foreign market than in the domestic market, then the forward discount for the foreign currency will **offset** the higher foreign interest rate*

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

1.

Covered and Uncovered interest rate parity

Covered interest rate parity (as demonstrated in LOS 1.c)

$$\begin{aligned}
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 &= S_{A/B} \times \frac{1+i_A - 1 - i_B}{1+i_B} \\
 &= S_{A/B} \times \frac{i_A - i_B}{1+i_B} \\
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Forward
premium/
discount

Covered interest rate parity

Interest
rate
differential

Note: Covered interest rate parity is **no-arbitrage** condition.
Disparity will be quickly arbitrated away $\rightarrow$ CIRP always holds

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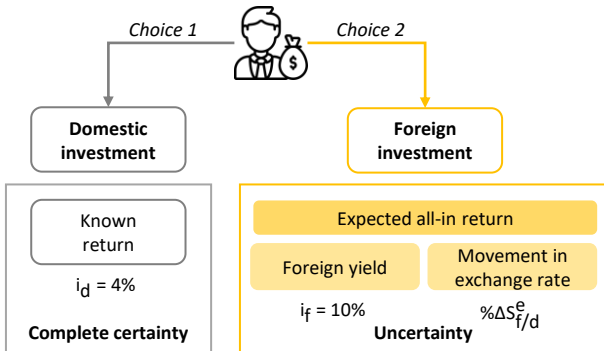
The Fisher effect, Real Interest Rate parity, and International Parity Conditions

1.

Covered and Uncovered interest rate parity

Uncovered interest rate parity

- Investor is assumed to transact at a future spot rate that is ***initially unknown***.
- The currency position in the future is not hedged – ***uncovered***.



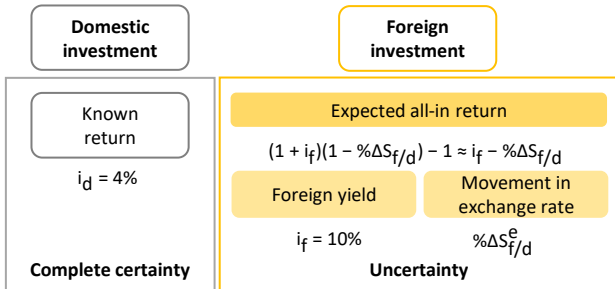
*For convenience, the notation for exchange rate will be foreign/domestic currency (f/d).

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

1.

Covered and Uncovered interest rate parity

Uncovered interest rate parity



Scenario 1

Spot rate is expected to remain unchanged $E(\% \Delta S) = 0$

Expected all-in foreign return = $10\% - 0 = 10\%$



Foreign investment

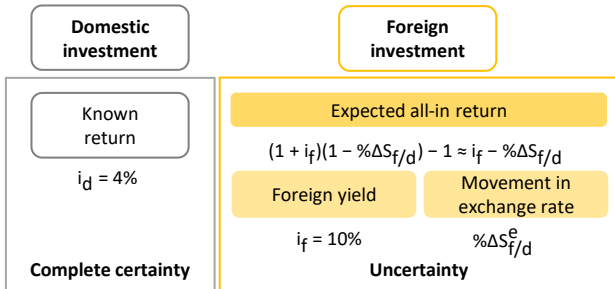
$i_d < \text{all-in foreign return}$

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

1.

Covered and Uncovered interest rate parity

Uncovered interest rate parity



Scenario 2

Domestic currency is expected to appreciate by
 $E(\% \Delta S) = 10\%$

Expected all-in foreign return = $10\% - 10\% = 0$



Domestic investment

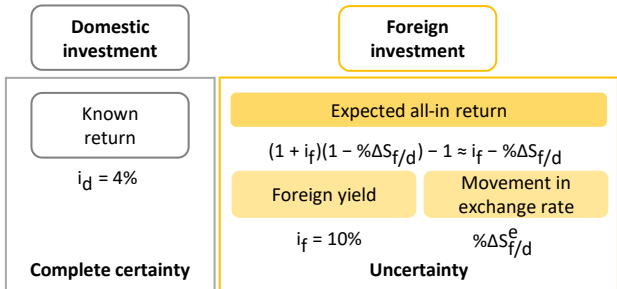
$i_d > \text{all-in foreign return}$

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

1.

Covered and Uncovered interest rate parity

Uncovered interest rate parity



Scenario 3

Domestic currency is expected to appreciate by
 $E(\% \Delta S) = 6\%$

Expected all-in foreign return = $10\% - 6\% = 4\%$



Indifferent

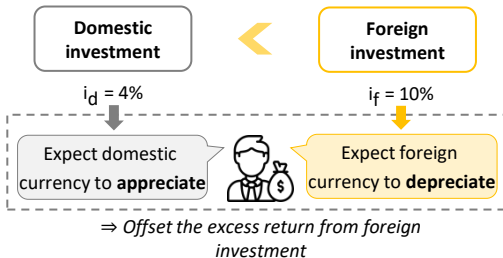
$i_d = \text{all-in foreign return}$

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

1.

Covered and Uncovered interest rate parity

Uncovered interest rate parity

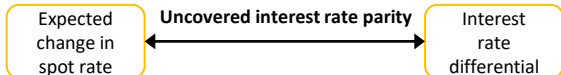


Uncovered: not bound by arbitrage

Uncovered interest rate parity

$$\% \Delta S_{f/d}^e = i_f - i_d$$

A risk neutral investor would be indifferent between both instruments because they offer the **same expected return**.



*For convenience, the notation for exchange rate will be foreign/domestic currency (f/d).

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

1.

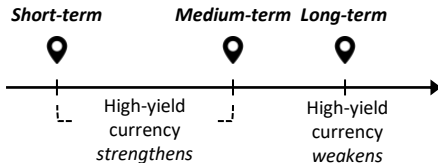
Covered and Uncovered interest rate parity

Uncovered interest rate parity

Uncovered interest rate parity in different time horizons

According to Uncovered interest parity: The country with *higher* interest rate will see its currency *depreciate*.

According to empirical studies:



- In short to medium term, high-yield currency tends to strengthen → UIRP is violated.
- In long-term, high-yield currency weakens → UIRP holds.

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

1.

Covered and Uncovered interest rate parity

Forward rate parity

Covered interest rate parity

Locked forward rate $F_{f/d}$

$$F_{A/B} = S_{A/B} \times \frac{1 + i_A}{1 + i_B}$$

$$\begin{aligned} F_{A/B} - S_{A/B} &= S_{A/B} \times \left(\frac{1 + i_A}{1 + i_B} - 1 \right) \\ &= S_{A/B} \times \frac{i_A - i_B}{1 + i_B} \text{ with } (1 + i_B \approx 1) \end{aligned}$$

$$\rightarrow F_{A/B} - S_{A/B} = S_{A/B} \times (i_A - i_B)$$

$$\rightarrow \frac{F_{A/B} - S_{A/B}}{S_{A/B}} = i_A - i_B$$

Uncovered interest rate parity

Unknown future spot rate

$$\% \Delta S_{A/B}^e = i_A - i_B$$



Forward rate parity

$$\frac{F_{A/B} - S_{A/B}}{S_{A/B}} = \% \Delta S_{A/B}^e$$

Forward
premium/
discount

Forward rate parity

Expected
change In spot
rate

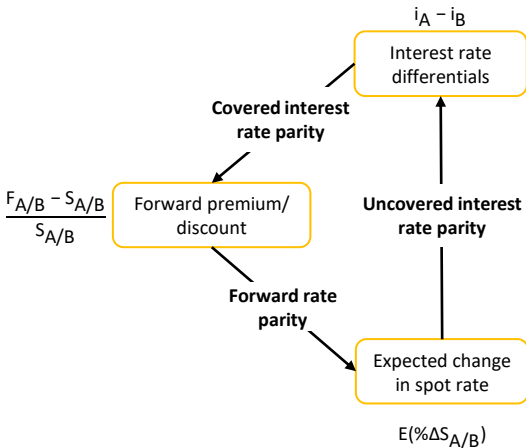
*Forward exchange rate is an **unbiased estimator** of the future spot exchange rate*

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

1.

Covered and Uncovered interest rate parity

Summary



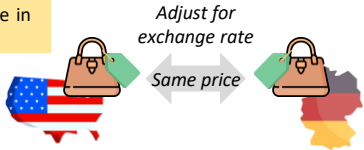
The Fisher effect, Real Interest Rate parity, and International Parity Conditions

2.

Purchasing power parity

Law of one price

Identical goods should have the same price in all locations



Absolute purchasing power parity

Law of one price (basket of goods)

Similar basket of goods should have the same price



Goods arbitrage will equate the prices of all goods and service across countries.

According to law of one price, the foreign price level of basket of goods (P_{EUR}) should equal the currency-adjusted domestic price level of the basket of goods (P_{USD}):

$$P_{EUR} = S_{EUR/USD} \times P_{USD}$$

$$\rightarrow S_{EUR/USD} = CPI_{EUR} / CPI_{USD}$$

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

2.

Purchasing power parity

Absolute purchasing power parity

Absolute PPP

$$S_{\text{EUR/USD}} = \text{CPI}_{\text{EUR}} / \text{CPI}_{\text{USD}}$$

The equilibrium exchange rate between two countries is determined entirely by the ratio of their national price levels.



Issues: Transaction costs are significant, not all goods and services are tradable.

→ Goods arbitrage will be incomplete and absolute PPP does not hold.

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

2. >> Purchasing power parity

Relative purchasing power parity

Assumption: Transaction costs and trade impediments are constant overtime.

$$\pi_{USD} = 4.9\%$$



$$\pi_{EUR} = 6.1\%$$



Recall: Fisher effect:

$$i = r + \pi$$

$$\pi_{USD} < \pi_{EUR} \rightarrow i_{USD} < i_{EUR}$$



Prefer investing
in i_{EUR}

Higher inflation leading to higher interest rate
causes i_{EUR} to be **more competitive** than i_{USD}

$S_{EUR/USD}$ **must rise** to maintain the competitiveness,
or EUR should **depreciate** relative to USD

Relative PPP

$$\% \Delta S_{EUR/USD} = \pi_{EUR} - \pi_{USD}$$

*The exchange rate changes to offset
changes in competitiveness arising
from inflation differentials.*

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

2. Purchasing power parity

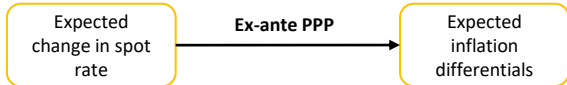
Ex-ante purchasing power parity

Comparing to relative PPP, **Ex-ante PPP** focus on expected changes in exchange rates driven by expected differences in inflation rates.

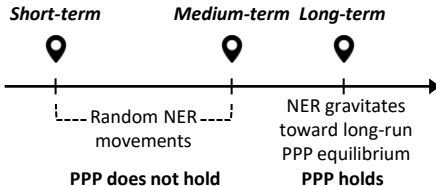
Ex-ante PPP

$$\% \Delta S_{\text{EUR/USD}}^e = \pi_{\text{EUR}}^e - \pi_{\text{USD}}^e$$

Countries that are **expected** to run persistently high inflation rates should **expect** to see their currencies depreciate over time, vice versa.



Purchasing power parity in different time horizons



The Fisher effect, Real Interest Rate parity, and International Parity Conditions

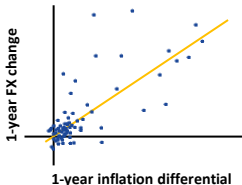
2. Purchasing power parity

Purchasing power parity in different time horizons

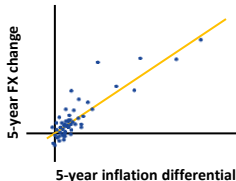
Effect of Relative Inflation Rates on Exchange Rates at Different Time Horizons:

As the time horizon is lengthened to five years and beyond, a strong positive relationship becomes apparent.

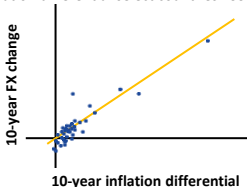
Relative PPP-1-year FX change vs. inflation differential selected countries 1990-2020



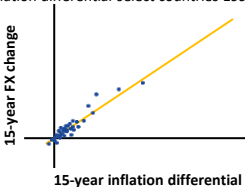
Relative PPP-5-year FX change vs. 5-year average inflation differential select countries 1990-2020



Relative PPP-10-year FX change vs. 10-year average inflation differential select countries 1990-2020



Relative PPP-15-year FX change vs. 15-year average inflation differential select countries 1990-2020

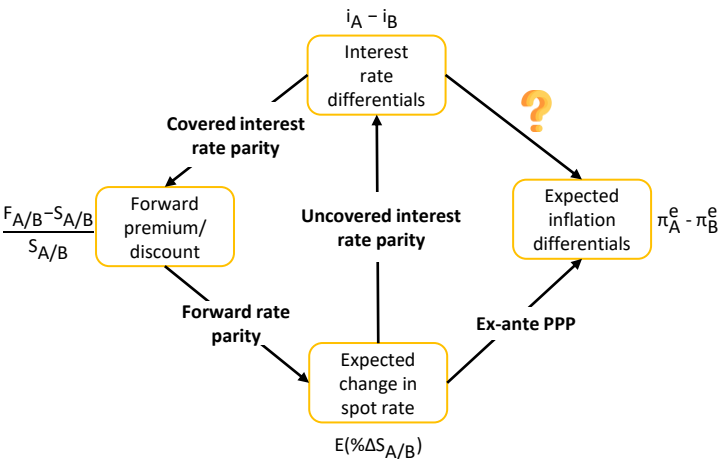


The Fisher effect, Real Interest Rate parity, and International Parity Conditions

2.

Purchasing power parity

Summary



The Fisher effect, Real Interest Rate parity, and International Parity Conditions

3.

International Fisher effect

Uncovered interest rate parity

$$\% \Delta S_{f/d}^e = i_f - i_d$$

Ex-ante PPP

$$\% \Delta S_{f/d}^e = \pi_f^e - \pi_d^e$$

$$\boxed{i_f - i_d} = \boxed{\pi_f^e - \pi_d^e}$$

Recall: Fisher effect

Nominal interest rate (i) = Real interest rate (r) + Expected inflation (π^e)

**Foreign
country**

$$i_f = r_f + \pi_f^e$$

**Domestic
country**

$$i_d = r_d + \pi_d^e$$

$$i_f - i_d$$

*Nominal
interest rate
differential*

$$=$$

$$r_f - r_d$$

*Real interest
rate
differential*

$$+$$

$$\pi_f^e - \pi_d^e$$

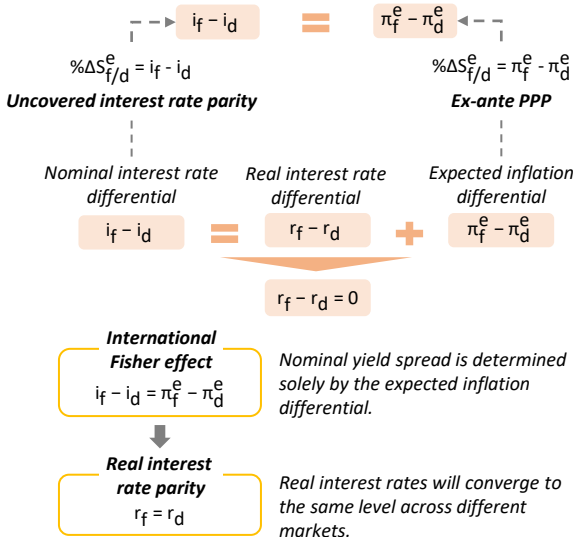
*Expected
inflation
differential*

**For convenience, the notation for exchange rate will be foreign/domestic currency (f/d).*

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

3.

International Fisher effect

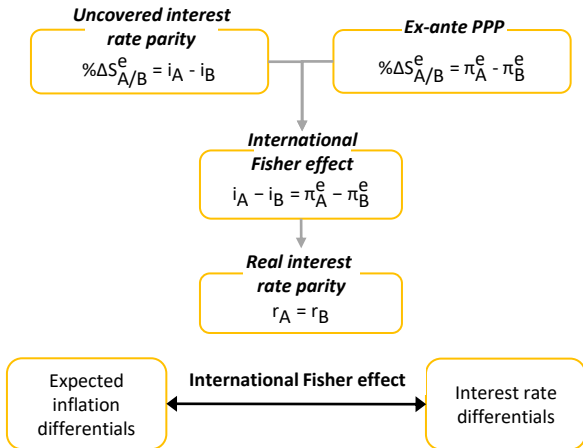


*For convenience, the notation for exchange rate will be foreign/domestic currency (f/d).

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

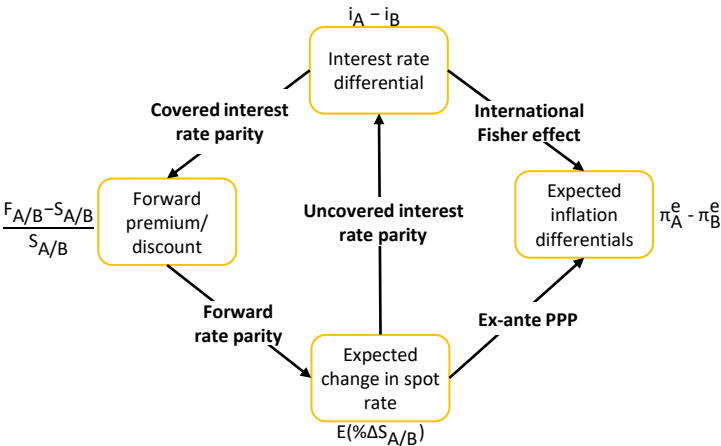
3.

International Fisher effect



The Fisher effect, Real Interest Rate parity, and International Parity Conditions

Summary



The Fisher effect, Real Interest Rate parity, and International Parity Conditions



Knowledge check

Example 1: Relationships among the International Parity Conditions

Forward rates are unbiased predictors of future spot rates if two parity conditions hold. Which of the following is not one of these conditions?

- A. Real interest rate parity
- B. Covered interest rate parity
- C. Uncovered interest rate parity

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

Knowledge check

Example 1: Relationships among the International Parity Conditions

Forward rates are unbiased predictors of future spot rates if two parity conditions hold. Which of the following is not one of these conditions?

- A. Real interest rate parity
- B. Covered interest rate parity
- C. Uncovered interest rate parity

Answer: A

Recall:

- *Covered interest rate parity:* $\frac{F_{A/B} - S_{A/B}}{S_{A/B}} = i_A - i_B$
- *Uncovered interest rate parity:* $\% \Delta S_{A/B}^e = i_A - i_B$

If the two parities hold, forward rate parity holds, which means forward rate is an unbiased predictor of the future spot rate:

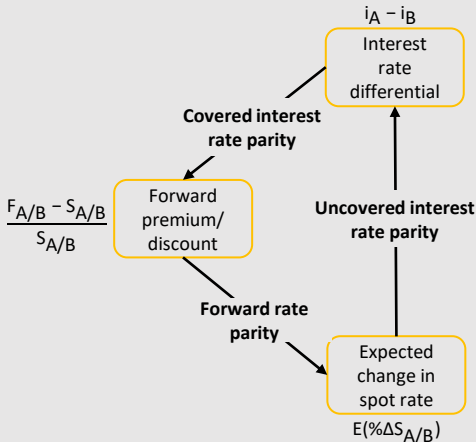
$$\frac{F_{A/B} - S_{A/B}}{S_{A/B}} = \% \Delta S_{A/B}^e$$

→ Real interest rate parity is not required.

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

Knowledge check

Example 1: Relationships among the International Parity Conditions



The Fisher effect, Real Interest Rate parity, and International Parity Conditions



Knowledge check

Example 2: Relationships among the International Parity Conditions

The international Fisher effect requires all but which of the following to hold?

- A. Ex ante PPP
- B. Absolute PPP
- C. Real interest rate parity

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

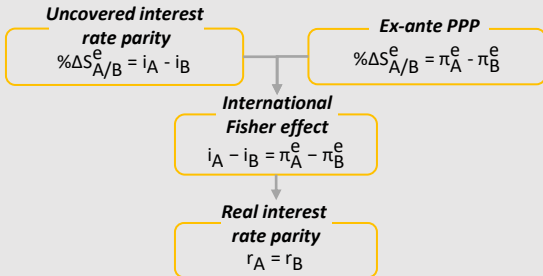
Knowledge check

Example 2: Relationships among the International Parity Conditions

The international Fisher effect requires all but which of the following to hold?

- A. Ex ante PPP
- B. Absolute PPP
- C. Real interest rate parity

Answer: B

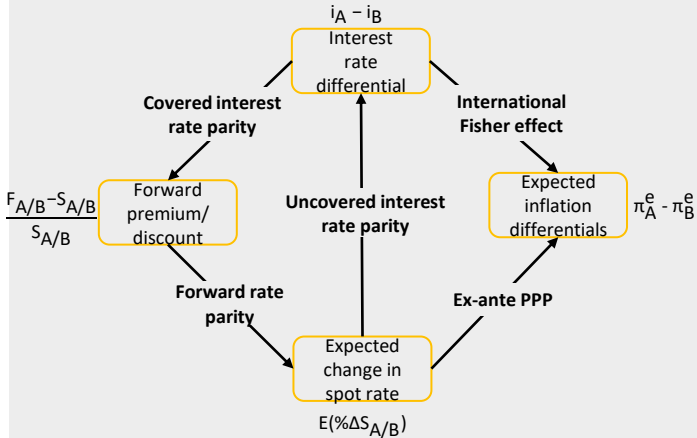


For International Fisher effect to hold, Ex ante PPP and Real interest rate parity must hold. Absolute PPP is not required.

The Fisher effect, Real Interest Rate parity, and International Parity Conditions

Knowledge check

Example 2: Relationships among the International Parity Conditions

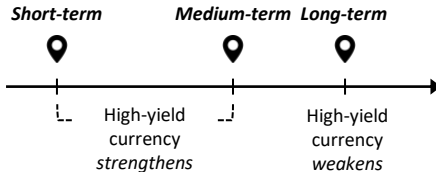


[LOS 1.i] Describe the carry trade and its relation to uncovered interest rate parity and calculate the profit from a carry trade

1.

Carry trade

Uncovered interest rate parity in different time horizons



UIRP is violated in short to medium term → There is an arbitrage opportunity.



Potential profitability of a trading strategy known as the **FX carry trade**



FX carry trade involves taking long positions in high-yield currencies and short positions in low-yield currencies.

[LOS 1.i] Describe the carry trade and its relation to uncovered interest rate parity and calculate the profit from a carry trade

1.

Carry trade

FX carry trade involves taking long positions in high-yield currencies and short positions in low-yield currencies.

Illustration:



$i_{CAD} = 1\%$

Also known as
Funding currency

**Low-yield
currency**



$i_{BRL} = 9\%$

**High-yield
currency**



FX carry trade

Step 1: Borrow CAD at 1%

Step 2: Convert to BRL at S_0

Step 3: Invest in BRL at 9%

Step 4: Convert back to CAD at S_1

Step 5: Payback the dollar loan

[LOS 1.i] Describe the carry trade and its relation to uncovered interest rate parity and calculate the profit from a carry trade

2. Profit from a carry trade

Illustration (cont.):



$i_{CAD} = 1\%$

**Low-yield
currency**



$i_{BRL} = 9\%$

**High-yield
currency**

$T = 0$

CAD 1

Convert to BRL at
spot rate $\times S_0$

BRL $(1 \times S_0)$

Payback loan
at $i_{CAD} = 1\%$

CAD $(1 + 1\%)$

Invest at
 $i_{BRL} = 9\%$

$T = 1$

CAD
 $S_0 \times (1 + 9\%) / S_1$

Convert to BRL
at spot rate S_1

BRL $S_0 \times (1 + 9\%)$

$$\text{Profit} = \frac{S_0 \times (1 + 9\%)}{S_1} - (1 + 1\%)$$

How does changes in spot rate affect the trader's profits?

[LOS 1.i] Describe the carry trade and its relation to uncovered interest rate parity and calculate the profit from a carry trade

2.

Profit from a carry trade

Illustration (cont.):



$i_{CAD} = 1\%$



$i_{BRL} = 9\%$

$$\text{Profit} = \frac{S_0 \times (1 + 9\%)}{S_1} - (1 + 1\%)$$

Carry trader's return = Intermarket yield spread (*known*)
+ Currency appreciation/depreciation (*unknown*)

1. BRL is expected to appreciate

Trader's profit will be greater than 8%

2. BRL is expected to depreciate less than 8%

Trader's profit will be less than 8%

3. BRL is expected to depreciate more than 8%

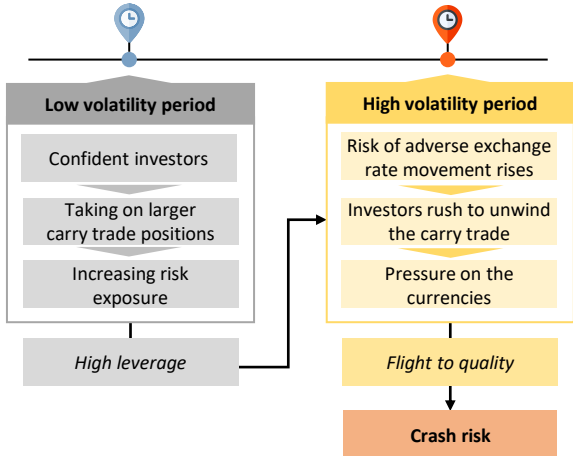
Trader will experience losses

[LOS 1.i] *Describe* the carry trade and its relation to uncovered interest rate parity and *calculate* the profit from a carry trade

3. Risk in carry trade

Reward of carry trade: gradual accrual of the interest rate differential – income unrelated to exchange rate volatility.

Risk of carry trade: sudden adverse exchange rate movements that result in instantaneous capital losses.



[LOS 1.i] *Describe* the carry trade and its relation to uncovered interest rate parity and *calculate* the profit from a carry trade

3.

Risk in carry trade

Return distribution of carry trade

Relative to a normal distribution, the distributions of carry trade return tend to be:

- More peaked around the mean → Larger number of trades with small gains/losses
- Fatter tails and negative skewness → More frequent and larger losses than normal indicating *Crash risk*.

[LOS 1.i] *Describe* the carry trade and its relation to uncovered interest rate parity and *calculate* the profit from a carry trade

 Knowledge check***Example 1:***

Carry trade can be profitable when:

- A. Covered interest rate parity does not hold.
- B. Uncovered interest rate parity does not hold.
- C. The international Fisher effect does not hold.

[LOS 1.i] Describe the carry trade and its relation to uncovered interest rate parity and calculate the profit from a carry trade

Knowledge check

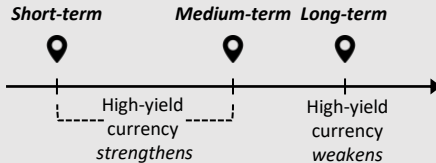
Example 1:

Carry trade can be profitable when:

- A. Covered interest rate parity does not hold.
- B. Uncovered interest rate parity does not hold.
- C. The international Fisher effect does not hold.

Answer: B

Uncovered interest rate parity in different time horizons



UIRP does not hold → Potential profitability of a trading strategy known as the **FX carry trade**

When UIRP holds, the change in spot rates over the tenor of the forward contracts would *cancel out* the interest rate differentials locked in at the inception of the position.

[LOS 1.i] *Describe* the carry trade and its relation to uncovered interest rate parity and *calculate* the profit from a carry trade

Knowledge check
Example 2:

A Tokyo-based asset manager enters into a carry trade position based on borrowing in yen and investing in one-year Australian MRR.

Today's One-year MRR		Currency pair	Spot rate today	Spot rate one year later
JPY	0.10%	JPY/USD	105.40	104.60
AUD	1.70%	USD/AUD	0.6810	0.6850

After one year, the all-in return to this trade, measured in JPY terms, would be *closest to*:

- A. +0.03%
- B. +1.43%
- C. +1.53%

[LOS 1.i] Describe the carry trade and its relation to uncovered interest rate parity and calculate the profit from a carry trade

Knowledge check

Answer: B

Because the carry trade involves 2 currencies JPY and AUD, but their exchange rates is not given, we first have to calculate the cross rate for this pair:

1. Determine the price and base currency of the required cross-rate, written as a fraction

$$\text{JPY/AUD} \rightarrow \frac{\text{JPY}}{\text{AUD}}$$

2. Determine the “third currency”

The third currency is USD, given the existing exchange rates.

3. Follow the fraction rules: *split the first fraction into two other fractions (exchange rates), so that the third currency can be crossed out*

$$\frac{\text{JPY}}{\text{AUD}} = \frac{\text{JPY}}{\cancel{\text{USD}}} \times \frac{\cancel{\text{USD}}}{\text{AUD}}$$

4. Calculate the cross rates

Current cross rate JPY/AUD = $105.40 \times 0.6810 = 71.78$

One-year cross rate JPY/AUD = $104.60 \times 0.6850 = 71.65$

5. Calculate the profit from the carry trade

(presented on the next slide)

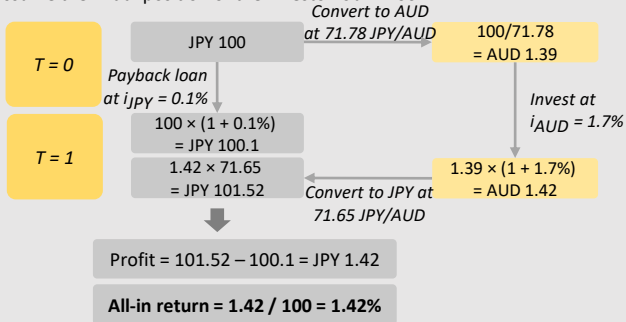
[LOS 1.i] Describe the carry trade and its relation to uncovered interest rate parity and calculate the profit from a carry trade

Knowledge check

Answer: B

5. Calculate the profit from the carry trade

Assume the initial position of the investor is JPY 100:

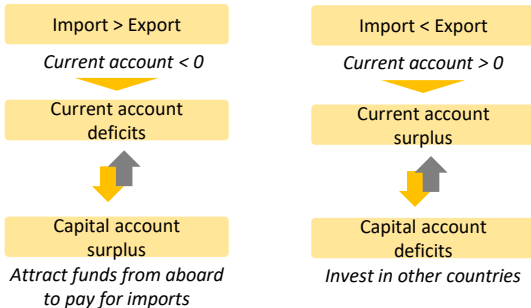
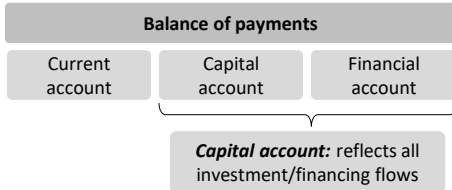


For quick calculation:

$$\text{All-in return} = \underbrace{\frac{1}{71.78}}_{\text{Convert to AUD}} \times \underbrace{(1 + 1.70\%)}_{\text{Invest at } i_{AUD}} \times \underbrace{71.65}_{\text{Convert back to JPY}} - \underbrace{(1 + 0.10\%)}_{\text{Repay loan at } i_{JPY} = 0.1\%} = 0.0142 = 1.42\%$$

[LOS 1.j] Explain how flows in the balance of payment accounts affect currency exchange rates

Recall: Balance of payments



[LOS 1.j] Explain how flows in the balance of payment accounts affect currency exchange rates

Current account: reflects flows in the real economy, actual production of goods and services

Capital account: reflects all investment/financing flows

Current account

Prices of real G&S tend to *adjust more slowly* than exchange rates and other asset prices.

Current spending/ production decisions reflect only purchases/sales of *current production*.

Production of G&S *takes time*, demand decisions are subject to substantial inertia.

Capital account

Expected exchange rate movements can induce very *large short-term capital flows*.

→ Actual exchange rate is very sensitive to the *currency views* held by owners/managers of liquid assets.

Investment/financing decisions reflect the financing of *current expenditures* and *reallocation of existing portfolios*.

Liquid financial markets allow virtually *instant redirection* of financial flows.

Financial flows have faster reaction, are the *dominant factor* in determining exchange rate movements (short to medium term).

[LOS 1.j] Explain how flows in the balance of payment accounts affect currency exchange rates

Summary: Effect of BOP on exchange rates

**Balance of
payment**

***1. Current account
imbalance***

The flow supply/demand channel (1.1)

Portfolio balance channel (1.2)

Debt sustainability channel (1.3)

2. Capital flows

[LOS 1.j] *Explain* how flows in the balance of payment accounts affect currency exchange rates

1.

Current Account Imbalances

1.1. The flow supply/demand channel

Purchases/sales of internationally traded G&S require the exchange of domestic and foreign currencies → Demand for currencies changes when exchange rate changes.

Trade surplus



Exchange for
domestic
currency ↑



Demand for
domestic
currency ↑



Domestic
currency
appreciates

Trade deficit



Exchange for
domestic
currency ↓



Demand for
domestic
currency ↓



Domestic
currency
depreciates

Can such trends go on indefinitely?

Strengthened
domestic
currency



Domestic price
increases



Competitiveness
deteriorates

Weakened
domestic
currency



Domestic price
decreases



Competitiveness
improves

*Persistent
surpluses/
deficits should
revert*

Exchange rate responses should help *eliminate the source of the initial imbalance* in the medium to long run.

[LOS 1.j] Explain how flows in the balance of payment accounts affect currency exchange rates

1.

Current Account Imbalances

1.1. The flow supply/demand channel

The *initial imbalance of current account will be eliminated* in the medium to long run by the exchange rate responses.

Amount of adjustment to restore balanced current accounts depends on:

1.

The initial gap between import and exports.

E.g.: Import exceeds export by a substantial amount → Require substantial depreciation

2.

The response of import and export prices to changes in the exchange rate: Limited pass-through effects of exchange rate changes on traded goods prices.

E.g.: Every 1% decline in a currency's value, import prices rise by only 5%

→ The exchange rate adjustment required to narrow a trade imbalance may be far larger

3.

The response of import/export demand to changes in import/export prices: The response demand to changes in price is quite sluggish → Relatively long lags

[LOS 1.j] Explain how flows in the balance of payment accounts affect currency exchange rates

1.

Current Account Imbalances

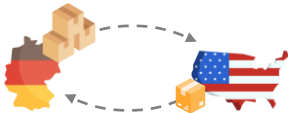
1.2. The Portfolio Balance Channel

***Recall:** Balance of payments – The current account deficit (surplus) will be offset by capital account surplus (deficit).*

Countries with trade deficits will finance their trade capital account surplus (by borrowing).

→ Investor countries (lenders) shift their global asset preferences (**Portfolio rebalancing**) → The path of exchange rates is influenced.

Example:



Germany runs large current account surplus vs. the US

Germany's holdings of USD-assets exceeds the desired amount

Germany **rebalance the portfolio:**
Reduces USD-assets to the desired amount

Demand for USD ↓

The value of USD ↓

[LOS 1.j] Explain how flows in the balance of payment accounts affect currency exchange rates

1.

Current Account Imbalances

1.3. The Debt Sustainability Channel

There should be upper limit on the ability of countries to run persistently large current account deficits.

**Import >
Export**

**Current
account
deficits**

**Capital
account
surplus**

**Debt owed
to foreign
investors ↑**

**Investor
confidence ↓**



**Currency
depreciation**

**Export ↑,
Import ↓**

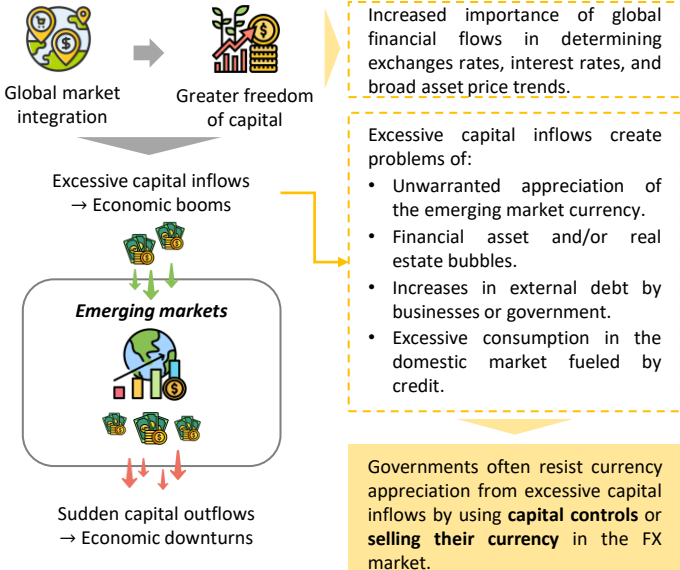
**Current account
rebalanced**

***External debt at
sustainable level***

[LOS 1.j] *Explain* how flows in the balance of payment accounts affect currency exchange rates

2.

Capital Flows



[LOS 1.j] *Explain* how flows in the balance of payment accounts affect currency exchange rates

2.

Capital Flows

Equity market trends and Exchange rates

Increasing equity prices can also attract foreign capital.

However, the relationship between equity market performance and exchange rates is not stable.

→ Difficult to form judgments on possible future currency movements based solely on expected equity market performance.

[LOS 1.k] *Explain the potential effects of monetary and fiscal policy on exchange rates*

Summary:
Effect of
Monetary
and fiscal
policies on
exchange
rates

1. Mundell-Flemming model

2. Monetary models

2.1. Pure monetary approach

2.2. Dornbusch overshooting model

3. Portfolio balance

[LOS 1.k] Explain the potential effects of monetary and fiscal policy on exchange rates

1.

The Mundell-Flemming Model

Mechanism:

Monetary and
fiscal policy

Interest rates
and economic
activity

Capital flows
and trade

Exchange rate

Capital flow mobility

High mobility (1.1)

Effect of monetary and fiscal policy on exchange rates will operate primarily through *capital flows*.

Low mobility (1.2)

Effect of monetary and fiscal policy on exchange rates will operate primarily through *trade flows*.

[LOS 1.k] *Explain the potential effects of monetary and fiscal policy on exchange rates*

1.

The Mundell-Flemming Model

1.1. Policy mix and Determination of Exchange rates under *High Capital Mobility*

Monetary policy	Expansionary	Interest rate ↓ ⇒ Capital outflow	Currency depreciation
	Contractionary	Interest rate ↑ ⇒ Capital inflow	Currency appreciation
Fiscal policy	Expansionary	Interest rate ↑ ⇒ Capital inflow	Currency appreciation
	Contractionary	Interest rate ↓ ⇒ Capital outflow	Currency depreciation

	Expansionary monetary policy	Restrictive monetary policy
Expansionary fiscal policy	Indeterminate	Domestic currency appreciates
Restrictive fiscal policy	Domestic currency depreciates	Indeterminate

[LOS 1.k] *Explain the potential effects of monetary and fiscal policy on exchange rates*

1.

The Mundell-Flemming Model

1.1. Policy mix and Determination of Exchange rates under *Low Capital Mobility*

Monetary policy	Expansionary	Import $\uparrow$ $\Rightarrow$ Trade deficit	Currency depreciation
	Contractionary	Import $\downarrow$ $\Rightarrow$ Trade surplus	Currency appreciation
Fiscal policy	Expansionary	Import $\uparrow$ $\Rightarrow$ Trade deficit	Currency depreciation
	Contractionary	Import $\downarrow$ $\Rightarrow$ Trade surplus	Currency appreciation

	Expansionary monetary policy	Restrictive monetary policy
Expansionary fiscal policy	Domestic currency depreciates	Indeterminate
Restrictive fiscal policy	Indeterminate	Domestic currency appreciates

[LOS 1.k] *Explain* the potential effects of monetary and fiscal policy on exchange rates

2.

Monetary Models of Exchange rate Determination

Mechanism: Output is fixed, and monetary policy affects exchange rates primarily through *the price level and the rate of inflation*.

Pure Monetary Approach

- Assumptions:**
- Prices are flexible.
 - Purchasing power parity holds in short and long runs.

Expansionary monetary policy → Domestic money supply increases

Domestic prices increase

Domestic price level > Foreign price level

Proportional decrease in the domestic currency's value

Shortcomings: PPP does not hold in the short or medium run, only holds in long run.

→ Pure monetary model *may not* provide a realistic explanation of the impact of monetary forces on the exchange rates.

[LOS 1.k] Explain the potential effects of monetary and fiscal policy on exchange rates

2.

Monetary Models of Exchange rate Determination

Dornbusch Overshooting Model

- Assumptions:**
- Prices have limited flexibility in the *short run* but fully flexible in the *long run*.
 - High mobility of capital.

In the short run

Expansionary monetary policy → Domestic money supply increases

Decline in domestic interest rate

Capital outflow

Domestic currency depreciate below long-run equilibrium

In the long run

Domestic MS ↑

Domestic prices ↑

Depreciation of domestic currency

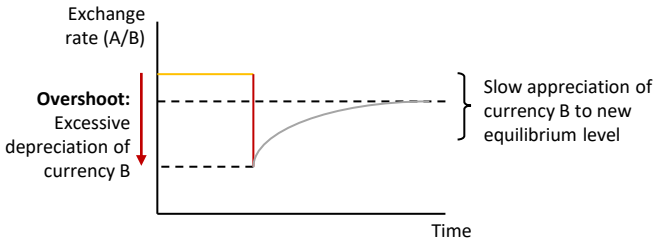
Consistent with the pure monetary model

[LOS 1.k] *Explain* the potential effects of monetary and fiscal policy on exchange rates

2.

Monetary Models of Exchange rate Determination

Dornbusch Overshooting Model



- In the short run, the equilibrium level will be reached through shifts in *financial market prices* (capital outflow).
- In the long run, the equilibrium level will be reached through shifts in the *price of goods themselves* (*similar to pure monetary approach*).

[LOS 1.k] Explain the potential effects of monetary and fiscal policy on exchange rates

3.

The Portfolio Balance Approach

Mechanism: Focuses only on the effects of *fiscal policy* and takes a *long-term view*, evaluates the effects of sustained fiscal deficit or surplus on currency values.

Assumption:

- Global investors are assumed to hold a diversified portfolio of domestic and foreign assets
- Desired allocation varies in response to changes in expected return and risk considerations.

Expansionary fiscal policy → Government budget deficit ↑

Building up debt → Unsustainable level of debt

Investors become skeptical on the unsustainable level of debt and refuse to fund the deficits

Currency depreciates

[LOS 1.k] *Explain* the potential effects of monetary and fiscal policy on exchange rates

4.

Integrated framework

The short- and long-run response of Exchange rates to changes in Fiscal policy

Expansive Fiscal policy

Short run response

Real interest rate
differential ↑

Currency appreciation

Mundell-Fleming model

Long run response

Government debt buildup

Central bank
monetizes debt

Fiscal stance turns
restrictive

Money supply ↑,
Interest rate ↓

Restrictive fiscal
policy, Interest
rate ↓

Currency depreciates

Portfolio balance model

[LOS 1.k] *Explain* the potential effects of monetary and fiscal policy on exchange rates



Knowledge check

Example: Fiscal policy and Exchange rates

Monique Kwan is doing her analysis of the foreign exchange rate outlook for selected countries. She examines a DM country that has a high degree of capital mobility and a floating-rate currency regime. Kwan notices that although the current outstanding volume of government debt is low, as a percentage of GDP, it is rising sharply as a result of expansionary fiscal policy. Moreover, projects for the government debt-to-GDP ratio point to further increases well into the future.

Kwan uses the Mundell-Fleming and portfolio balance models to form an opinion about both the short-run and long-run implications for the DM country's exchange rate.

Question 1: Over the short run, Kwan is *most likely* to expect:

- A. Appreciation of the DM currency.
- B. An increase in the DM's asset prices.
- C. A decrease in the DM's risk premium.

[LOS 1.k] Explain the potential effects of monetary and fiscal policy on exchange rates

Knowledge check

Question 1: Over the short run, Kwan is *most likely* to expect:

- A. Appreciation of the DM currency.
- B. An increase in the DM's asset prices.
- C. A decrease in the DM's risk premium.

Answer: A

DM country situation:

- Low volume of outstanding government debt
- High capital mobility
- Floating-rate currency regime

The question asks for the **short-run response**

→ Follow the Mundell-Fleming model → Appreciation of the DM's currency

Expansive Fiscal policy

Short run response

Real interest rate differential ↑

Currency appreciation

Mundell-Fleming model

Long run response

Government debt buildup

Central bank
monetizes debt

Fiscal stance
turns restrictive

Currency depreciates

Portfolio balance model

[LOS 1.k] *Explain* the potential effects of monetary and fiscal policy on exchange rates



Knowledge check

Example: Fiscal policy and Exchange rates

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Kwan uses the Mundell-Fleming and portfolio balance models to form an opinion about both the short-run and long-run implications for the DM country's exchange rate.

Question 2: Over the medium term, as the DM country's government debt becomes harder to finance, Kwan would be *most likely* to expect that:

- A. Fiscal policy will turn more accommodative.
- B. The mark-to-market value of the debt will increase.
- C. Monetary policy will become more accommodative.

[LOS 1.k] *Explain* the potential effects of monetary and fiscal policy on exchange rates

Knowledge check

Answer: C

The question asks for the **medium-term response** → Follow the Portfolio balance model

DM country's situation: Government debt is rising sharply → Government debt buildup

→ Government will be tempted to monetize the debt through an accommodative monetary policy (expansionary monetary policy).

Expansive Fiscal policy

Short run response

Real interest rate differential ↑

Currency appreciation

Mundell-Fleming model

Long run response

Government debt buildup

Central bank
monetizes debt

Fiscal stance
turns restrictive

Currency depreciates

Portfolio balance model

[LOS 1.k] *Explain* the potential effects of monetary and fiscal policy on exchange rates



Knowledge check

Example: Fiscal policy and Exchange rates

Monique Kwan is doing her analysis of the foreign exchange rate outlook for selected countries. She examines a DM country that has a high degree of capital mobility and a floating-rate currency regime. Kwan notices that although the current outstanding volume of government debt is low, as a percentage of GDP, it is rising sharply as a result of expansionary fiscal policy. Moreover, projects for the government debt-to-GDP ratio point to further increases well into the future.

Kwan uses the Mundell-Fleming and portfolio balance models to form an opinion about both the short-run and long-run implications for the DM country's exchange rate.

Question 3: Assuming that the DM's country's government debt becomes harder to finance and there is no change in monetary policy, Kwan is most likely to expect that over the longer term, there will be a fiscal policy response that will lead to:

- A. Currency appreciation as yields rise.
- B. Currency depreciation as yields decline.
- C. An indeterminate impact on the currency, depending on which effect prevails.

[LOS 1.k] Explain the potential effects of monetary and fiscal policy on exchange rates

Knowledge check

Answer: B

DM country's situation:

- Government debt becomes harder to finance → Government debt buildup (*Follow the Portfolio balance model*)
 - No change in monetary policy → Central bank will not monetize the rising debt
- The most likely fiscal response is an eventual move toward fiscal consolidation – reducing the public deficit and debt levels → Issue fewer bonds.
- Bond yields would decrease → Weaker domestic currency in the long-run.

Expansive Fiscal policy

Short run response

Real interest rate differential ↑

Currency appreciation

Mundell-Fleming model

Long run response

Government debt buildup

Central bank
monetizes debt

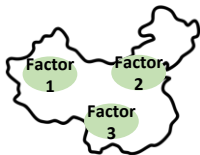
Fiscal stance
turns restrictive

Currency depreciates

Portfolio balance model

[LOS 1.I] *Describe objectives of central bank or government intervention and capital controls and describe the effectiveness of intervention and capital controls*

Capital flows drivers

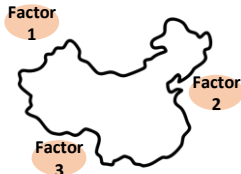


Pull factors

A favorable set of developments that encourage foreign capital inflows.

E.g.:

- *Price stability*
- *Flexible exchange rate regime*
- *Improved fiscal position*
- *Privatization of SOEs*



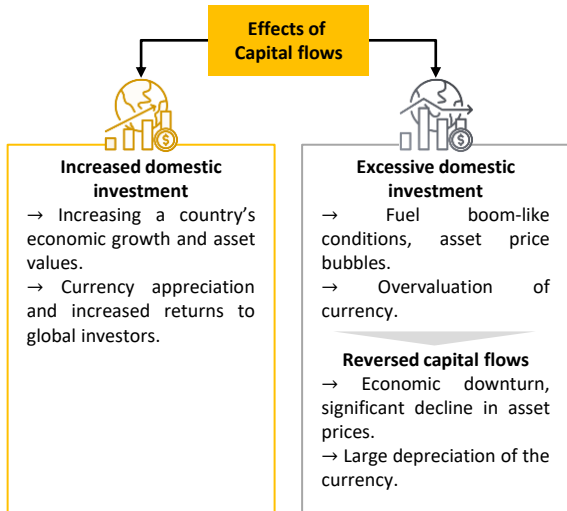
Push factors

Are largely driven by mobile international capital seeking high returns from a diversified portfolio.

E.g.: Low interest rate policies in industrial countries since 2008 financial crisis

→ Encouraged global investors to seek higher returns abroad.

[LOS 1.I] Describe objectives of central bank or government intervention and capital controls and describe the effectiveness of intervention and capital controls



Policymakers must guard against excessive capital inflows that could quickly be reversed

[LOS 1.I] *Describe objectives of central bank or government intervention and capital controls and describe the effectiveness of intervention and capital controls*

1.

Objective of government intervention and capital controls

- Ensure that the domestic currency does not appreciate excessively.
- Allow the pursuit of independent monetary policies without being hindered by their impact on currency values.

E.g.: Emerging market central bank seeks to reduce inflation → Restrictive monetary policy → Increasing interest rates → Attract inflows of foreign capital → Upward pressure on currency value

- Reduce the aggregate volume of inflow of foreign capital.

2.

Effectiveness of government intervention and capital controls

- **Developed market economies:**

Volume of FX trading > Central bank FX reserves

→ Insufficient reserves to significantly affect the supply and demand for domestic currency.

→ *Limited effectiveness.*

- **Emerging market economies:**

Daily FX trading < Central bank FX reserves

→ Sufficient reserves to affect the supply and demand for domestic currency.

→ *Greater success in controlling exchange rates.*

[LOS 1.m] *Describe* warning signs of a currency crisis

Typical warning signs of a currency crisis

Capital markets	Prior to a crisis, capital market been liberalized to allow the free flow of capital.
Foreign capital	Large inflows in the period leading up to a crisis, with short-term funding denominated in a foreign currencies.
Banking crisis	Currency crises are often preceded by (and often coincide with) banking crisis.
Exchange rate regimes	Countries with fixed or partially fixed exchange rates are more susceptible to currency crisis.
Foreign reserves	Foreign exchange reserves tend to decline sharply before a crisis.
Currency value	The currency has risen substantially relative to its historical mean.
Terms of trade	The ratio of exports to imports often deteriorates before a crisis.
Broad money	Broad money growth and the ratio of M2 to bank reserves tend to rise prior to a crisis.
Inflation	Inflation tends to be significantly higher in pre-crisis periods compared with tranquil periods.



LEARNING OUTCOMES

- 2.a.** Compare factors favoring and limiting economic growth in developed and developing economies
- 2.b.** Describe the relation between the long-run rate of stock market appreciation and the sustainable growth rate of the economy
- 2.c.** Explain why potential GDP and its growth rate matter for equity and fixed income investors
- 2.d.** Contrast capital deepening investment and technological progress and explain how each affects economic growth and labor productivity
- 2.e.** Demonstrate forecasting potential GDP based on growth accounting relations
- 2.f.** Explain how natural resources affect economic growth and evaluate the argument that limited availability of natural resources constrains economic growth
- 2.g.** Explain how demographics, immigration, and labor force participation affect the rate and sustainability of economic growth

LEARNING OUTCOMES

- 2.h.** Explain how investment in physical capital, human capital, and technological development affects economic growth
- 2.i.** Compare classical growth theory, neoclassical growth theory, and endogenous growth theory
- 2.j.** Explain and evaluate convergence hypotheses
- 2.k.** Describe the economic rationale for governments to provide incentives to private investment in technology and knowledge
- 2.l.** Describe the expected impact of removing trade barriers on capital investment and profits, employment and wages, and growth in the economies involved

[LOS 2.a] *Compare* factors favoring and limiting economic growth in developed and developing economies

1.

Savings and Investment



Role of Savings and Investment

Savings

Investments

GDP growth

 Contribution to
Incomes


Problems of developing countries

Vicious cycle of poverty:

Low savings → Low levels of investments → Slow
GDP growth → Persistently low income and savings

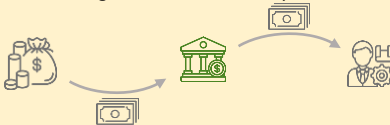
Typically aimed remedies: Capital growth & FDI

[LOS 2.a] *Compare* factors favoring and limiting economic growth in developed and developing economies

2.

Financial Markets and Intermediaries

A role of the **financial sector** in any economy is to channel funds from savers to investment projects → Saving is allocated efficiently within the economy.



Screening those who seek funding and monitoring those who obtain funding → Channels savings to projects with highest risk-adjusted returns.



Creating attractive investment instruments
→ Facilitate risk transfer and diversification and enhance liquidity



Aggregate small savings into larger pools of funds to finance larger projects
→ Mitigating credit constraints that companies might face

**Promote
economic
growth
by:**

[LOS 2.a] *Compare* factors favoring and limiting economic growth in developed and developing economies

3.

Political stability, Rule of law, and Property rights



Political stability

Stable and effective government



Legal system

Needed to establish and protect property rights



Property rights

Legal arrangements that govern the protection of private property

Encourage foreign investment and strengthen growth.

- **Substantive law:** rights and responsibilities of entities and relationships among entities.

- **Procedure law:** focuses on the protection and enforcement of the substantive laws.

Incentive for domestic households and companies to invest and save.

- **Developed countries:** Rights and arrangements are well established and effective.
- **Developing countries:**
 - May be lacking rights and arrangements or be ineffective.
 - Economic uncertainty increases investment risk, discourage investment and weaken growth.

[LOS 2.a] Compare factors favoring and limiting economic growth in developed and developing economies

4.

Education and Health care systems



Education system

Basic education raises the skill level of the workforce and contributes to the country's potential growth



Problems of developing countries

- Illiterate and poorly skilled workers
- *"Brain drain"*: most highly educated individuals leave the developing country for more-advanced countries



Health care system

Healthy labor with high life expectancy rates means they are available to work → Foreign investors are more willing to invest

5.

Tax and Regulatory Systems

- Lower tax and regulatory burdens encourage entrepreneurial activities and the entry of new companies.
 - Entry of new companies is strongly positively correlated with average productivity levels.
- Boost growth and productivity, especially at company level.

[LOS 2.a] *Compare* factors favoring and limiting economic growth in developed and developing economies

6.

Free trade and Unrestricted capital flows



Free trade

- Providing residents with more goods at lower costs
- Domestic companies face increased competition but obtain access to larger markets.



Unrestricted capital flows

- Invest directly in domestic economy (FDI)
- Invest indirectly in domestic economy by purchasing securities



Developing countries can **break the cycle** of low income, low domestic savings and low investment (*refer to Problems of developing country on slide 122*) through **FDI**:

- Increase physical capital stock
- Higher productivity, employment, wages, savings

[LOS 2.a] *Compare* factors favoring and limiting economic growth in developed and developing economies

Factors	Comparison	
	<i>Developed countries</i>	<i>Developing countries</i>
Savings and Investments	Higher level of savings and investment	Low rates of savings and investment → Cycle of poverty
Financial market and Intermediaries	Well developed financial markets	Poorly developed financial markets
Political stability, Rule of law, Property rights	Well established rights and arrangements	• Lacking rights and arrangements • Ineffective legal system
Education and Health care systems	• On the leading edge of technology • High life expectancy rates	• Poorer public education • Poor health services
Free trade and Unrestricted capital flows	• Encourage investments abroad • Face higher competition • Access to larger market	• Benefit from FDIs → Break the cycle of poverty • Face higher competition • Access to larger markets

[LOS 2.b,c] Describe the relation between the long-run rate of stock market appreciation and the sustainable growth rate of the economy; Explain why potential GDP and its growth rate matter for equity and fixed income investors

1. Equity versus Economy

Potential GDP

- Measures growth in an economy's productive capacity
- Places a limit on how fast the economy can grow

→ Potential GDP is the maximum amount of output an economy can sustainably produce without inducing an increase in the inflation rate.

Can earnings growth exceed growth rate of potential GDP?

$\frac{\text{Corporate profits}}{\text{GDP}}$ ↑



Earnings growth



Potential GDP growth

Problems: Stagnant labor income would make workers unwilling to work and would also undermine demand → **Unsustainable further profit growth.**

In the long run, earnings growth is limited by potential GDP growth

[LOS 2.b,c] Describe the relation between the long-run rate of stock market appreciation and the sustainable growth rate...

1.

Equity versus Economy

Grinold-Kroner decomposition of return to equity

$$E(R_e) = \text{Dividend yield} + \text{Expected capital gain}$$

$$E(R_e) = \text{Dividend yield} + \text{Expected repricing} + \text{Earnings growth per share}$$

$$E(R_e) = \text{Dividend yield (dy)} + \text{Expected repricing } \Delta(P/E) + \text{Inflation rate (i)} + \text{Real economic growth (g)} + \text{Change in shares outstanding } (\Delta S)$$

→ According to Grinold-Kroner decomposition, economic growth is one of the components of the return on equities.

[LOS 2.b,c] Describe the relation between the long-run rate of stock market appreciation and the sustainable growth rate...

1.

Equity versus Economy

Grinold-Kroner decomposition of return to equity

$$E(R_e) = \text{Dividend yield (dy)} + \text{Expected repricing } \Delta(P/E) + \text{Inflation rate (i)} + \text{Real economic growth (g)} + \text{Change in shares outstanding } (\Delta S)$$

Dividend yield (dy): Fairly stable and a significant contributor to equity market returns

Expected repricing $\Delta(P/E)$: Increasing GDP growth rate $\rightarrow$ P/E ratios trend higher as investors perceive the country and its economy to be less risky $\rightarrow$ Investor willing to pay a higher price for earnings.

Earning growth per share $(i + g + \Delta S)$: Primary channel through which economic growth can impact equity returns

Change in shares outstanding (ΔS) - Dilution effect: Plays an important role in determining expected equity returns, arises due to:

- *Publicly traded companies* issuing new shares and dilute existing shareholders' return or buy back shares and increase equity return $\rightarrow$ **Net buybacks (nbb)**.
- *Private small and medium-sized companies* possessing greater economic growth $\rightarrow$ **Relative dynamism (rd)**.

$$\rightarrow \Delta S = nbb + rd$$

[LOS 2.b,c] Describe the relation between the long-run rate of stock market appreciation and the sustainable growth rate...

2.

Fixed Income versus Economy

How potential GDP forecasts affect fixed income market

Channel	Mechanism	Effect on fixed income market
<i>Inflationary pressure</i>	Actual GDP growth > Potential GDP growth → Upward pressure on inflation	Nominal interest rates increase, and bond prices decrease
<i>Real interest rates</i>	Fast potential GDP growth → Expected rise in income → Require higher real interest rate to induce savings	Higher real interest rates and higher expected real asset returns
Other factors	Effect on fixed income market	
<i>Credit quality</i>	Higher rate of potential GDP growth improves the general credit quality of fixed-income securities.	
<i>Monetary policy</i>	Output gap ($Y - Y^*$) and actual GDP growth relative to long-term sustainable growth rate affect monetary policy of the central bank	Affect interest rate and economic activities in general.
<i>Fiscal policy</i>	Actual fiscal positions are judged relative to structural or cyclical adjusted deficits (budget of an economy operating at <i>its potential GDP</i>).	

[LOS 2.d,e] Contrast capital deepening investment and technological progress and *explain* how each affects economic growth and labor productivity; *Demonstrate* forecasting potential GDP based on growth accounting relations

1.

Production function

Simple production function

$$Y = A \times F(K, L)$$

Where:

- Y = The level of aggregate output in the economy
- L = The quantity of labor / number of workers / number of hours worked
- K = Estimate of the capital from equipment and structures
- A = Total factor productivity (TFP)

Where:

- α = the *share of GDP* paid out to the suppliers of *capital*.
- $1 - \alpha$ = the *share of GDP* paid to *labor*.

Cobb-Douglas production function

$$Y = AF(K, L) = AK^\alpha L^{1-\alpha}$$

Factors of production are paid their marginal product

Marginal product of capital (MPK):

$$\begin{aligned} \text{MPK} &= \alpha AK^{\alpha-1} L^{1-\alpha} \\ &= \alpha \times Y/K \end{aligned}$$

Profit maximization requires: Marginal product of capital (MPK) = rental price of capital (r)

$$\rightarrow \text{MPK} = \alpha \times Y/K = r \rightarrow \alpha = rK/Y$$

where:

- α is the *share of GDP* paid out to the suppliers of *capital*.
- $1 - \alpha$ is the *share of GDP* paid to *labor*.

[LOS 2.d,e] Contrast capital deepening investment and technological progress and *explain* how each affects economic growth ...

1.

Production function

Constant returns to scale

If all the inputs into the production process are increased by the same percentage, then output rises by that percentage.

Quantity of output per worker

Production function

$$Y/L = AF(K/L, L/L) = AF(K/L, 1)$$

$y = Y/L$ = output per worker or labor productivity

$k = K/L$ = capital to labor ratio

$$\rightarrow y = AF(k, 1)$$

Cobb-Douglas production function

$$y = Y/L = A(K/L)^\alpha (L/L)^{1-\alpha} = Ak^\alpha$$

Amount of goods a worker can produce (y) depends on:

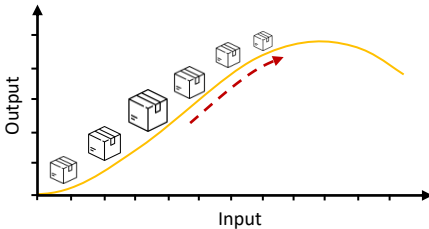
- Amount of capital available for each worker (k)
- Technology or TFP (A)
- The share of capital in GDP (α)

[LOS 2.d,e] **Contrast** capital deepening investment and technological progress and *explain* how each affects economic growth ...

1.

Production function

Diminishing marginal productivity



The **extra output** obtained from each **additional unit of input** will decline (*output grows slower*).

→ Each additional worker adds less to output than previously added worker, average labor productivity (y) falls.

Cobb-Douglas production function

$$y = Y/L = A(K/L)^\alpha (L/L)^{1-\alpha} = Ak^\alpha$$

- α close to zero → Diminishing marginal returns to capital are *very significant*.
- α close to one → Diminishing marginal returns has relatively *small impact*.

[LOS 2.d,e] Contrast capital deepening investment and technological progress and *explain* how each affects economic growth ...

2.

Growth accounting

Growth accounting equation (Solow model)

 $\Delta Y/Y$
 $=$
 $\Delta A/A$
 $+$
 $\alpha \Delta K/K$
 $+$
 $(1 - \alpha) \Delta L/L$

*Growth
rate of
output*

*Rate of
technology
change*

*Growth rate
of capital
($\Delta K/K$)*

*Growth rate
of labor ($\Delta L/L$)*

Where:

- α = the elasticity of output with respect to *capital*
- $1 - \alpha$ = the elasticity of output with respect to *labor*

Uses of growth accounting equation

- Estimate the contribution of technological progress: Solow estimated the growth in TFP as *residual* (the amount of output that cannot be explained by growth in capital or labor).
- Measure the sources of growth: Either from *growth of capital* ($\Delta K/K$) or *growth of labor* ($\Delta L/L$).
- Measure potential output (*will be mentioned in the next slide*).

[LOS 2.d,e] Contrast capital deepening investment and technological progress and *explain* how each affects economic growth ...

2.

Growth accounting

Methods to Measure potential GDP

1. Solow growth accounting equation

$\Delta Y/Y$	=	$\Delta A/A$	+	$\alpha \Delta K/K$	+	$(1 - \alpha) \Delta L/L$
<i>Growth rate of output</i>		<i>Rate of technology change</i>		<i>Growth rate of capital ($\Delta K/K$)</i>		<i>Growth rate of labor ($\Delta L/L$)</i>

2. Labor productivity growth accounting equation

Growth rate of potential GDP	=	Long-term growth rate of labor force	+	Long-term growth rate in labor productivity
------------------------------	---	--------------------------------------	---	---

Advantages:

- Simpler model: function of labor and productivity of labor input.
- Avoids the need to estimate capital input and total factor productivity.

Disadvantage: Incorporates capital deepening and TFP progress in productivity term

→ Difficult to analyze and predict over long periods.

[LOS 2.d,e] Contrast capital deepening investment and technological progress and *explain* how each affects economic growth ...

2.

Growth accounting

Extending the Production function

A more complete specification of the production function expands the list of inputs:

- Raw material (N) (*LOS 2.f*)
- Quantity of labor (L) (*LOS 2.g*)
- Human capital (H) (*LOS 2.g*)
- Information, computer, and telecommunication (ICT) capital (KIT) (*LOS 2.h*)
- Non-ICT capital (KNT) (*LOS 2.h*)
- Public capital (KP) (*LOS 2.h*)
- Technological knowledge (KIT) (*LOS 2.h*)

$$Y = AF(N, L, H, KIT, KNT, KP)$$

[LOS 2.d,e] Contrast capital deepening investment and technological progress and *explain* how each affects economic growth ...

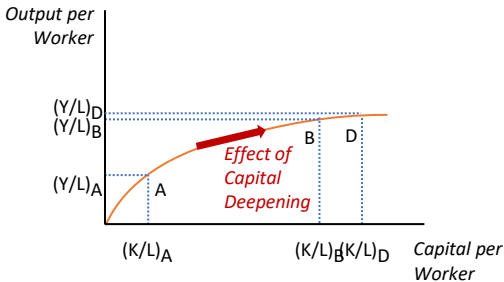
3.

Capital deepening versus Technological progress

According to **diminishing marginal returns**, marginal product of capital declines as more capital is added to the labor input

Capital deepening

- Rising investment in the economy leads to an increase in the capital-to-labor ratio → A move long the production function from point A to point B.
- Further additions have relatively little impact on per capita output → A move long the production function from point B to point D.



[LOS 2.d,e] **Contrast** capital deepening investment and technological progress and *explain* how each affects economic growth ...

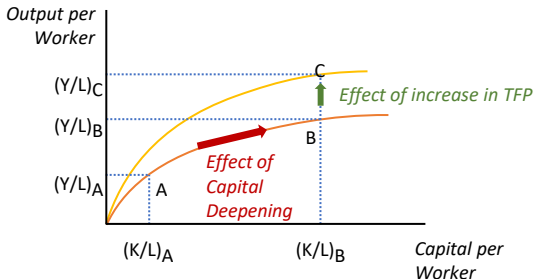
3.

Capital deepening versus Technological progress

Technological progress: New and innovative products and processes

Higher output per worker (Y/L) given a level of capital (K/L) → Point B to point C

Upward shift in the entire function



Technological progress mitigates the limits of diminishing marginal returns
 → Sustained growth in per capita output requires progress in TFP.

[LOS 2.f] *Explain* how natural resources affect economic growth and *evaluate* the argument that limited availability of natural resources constrains economic growth

Natural resources

Renewable resources



Resources that can be replenished.

E.g.: Trees can be cut, and seedling can be planted into a new forest.

Non-renewable resources



Finite resources that are depleted once they are consumed.

E.g.: Coal and oil.



Are countries with more natural resources wealthier?

[LOS 2.f] Explain how natural resources affect economic growth and evaluate the argument that limited availability...

Are countries with more natural resources wealthier?

- Ownership and production of natural resources is not necessary for a country to achieve a high level of income.

E.g.: South Korea have experienced rapid economic growth with only few natural resources.

Venezuela have large oil reserves and is one of the major producers of oil, but experienced subpar growth.

- Presence of natural resources may restrain growth → **Resource curse**



Countries rich in resources may fail to develop the *economic institutions* necessary for growth.



Dutch disease: Currency appreciation driven by strong export demand for resources makes other segments of the economy uncompetitive.

[LOS 2.g] *Explain* how demographics, immigration, and labor force participation affect the rate and sustainability of economic growth

Growth in labor input

1. Labor Quantity

1.1. Population Growth

- Determined by fertility rates and mortality rates.
- Growth rate of developed countries population < Growth rate of developing countries population
- Age mix (*the percentage of population in different age groups*) is also important.

1.2. Labor Force Participation

$$\text{Labor force participation} = \frac{\text{Labor force}}{\text{Working age population}}$$

- May make labor force growth rate differ from population growth.
- Increase in the participation rate may raise the growth of per capita GDP.

1.3. Net Migration

Heightened immigration is a possible solution to the slowing labor force growth of developed countries.

1.4. Average Hours Worked

- Affect the contribution of labor to overall output, highly sensitive to the business cycle.
- Long-term trends toward a shorter work week in developed countries.

2. Labor Quality

Human capital: Accumulated knowledge and skills that workers acquire → Productivity and adaptability to changes, promote innovation.

[LOS 2.h] Explain how investment in physical capital, human capital, and technological development affects economic growth

1.

Physical capital

Investment



Depreciation



Physical capital
stock increases

Countries with higher rate of investment should have a growing physical capital stock and a higher rate of GDP growth.

[LOS 2.h] *Explain* how investment in physical capital, human capital, and technological development affects economic growth

1.

Physical capital



If long-term sustainable growth cannot rely on pure capital deepening (*LOS 2.d,e*), how can investment and economic growth be highly correlated?

1

Investment-driven growth may last for a considerable period, especially countries with relatively low levels of capital per worker

2

Investment spending on available capital depends on the existing physical capital stock

3

Composition of investment spending and the stock of physical capital matters for growth and productivity

[LOS 2.h] **Explain** how investment in physical capital, human capital, and technological development affects economic growth

1.

Physical capital

1. Investment-driven growth may last for a considerable period
2. Investment spending on available capital depends on the existing physical capital stock

Developing country

Low level of physical capital per worker

→ The *positive impact* of changes in the physical capital stock on growth is *very significant*.

Developed country

High level of physical capital per worker

→ Need a *sustained high level* of investment over many years to have a *meaningful relative impact* on the physical capital stock.

3. Composition of investment spending and the stock of physical capital matters for growth and productivity

ICT investment

- **Includes:** Information, computers, telecommunications equipment.
- High levels of ICT investments (*increase TFP*) may *actually boost* the growth rate of *potential GDP*.

Non-ICT investment

- **Includes:** Non-residential construction, transport equipment, machinery.
- High levels of non-ICT capital spending should eventually result in *capital deepening*.

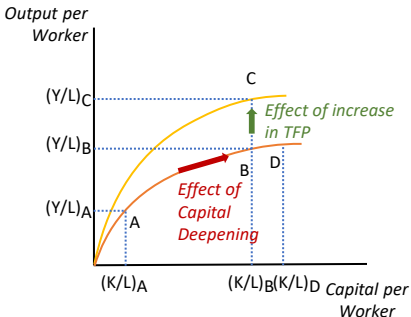
[LOS 2.h] Explain how investment in physical capital, human capital, and technological development affects economic growth

2.

Technological development

Technology

- The most important factor affecting growth of per capita GDP, especially in developed countries.
- Changes can be **embodied** in **human capital** and/or new machinery, equipment and software.



Technology allows an economy to overcome some of the limits imposed by diminishing marginal returns

Produce more/higher-quality goods and services with the same resources or inputs

Creation of new goods and services

Efficiently organized and managed businesses

[LOS 2.h] Explain how investment in physical capital, human capital, and technological development affects economic growth

3.

Public infrastructure

- The definition of the capital input also includes public infrastructure (e.g., roads, bridges, municipal water, dams, electric grids,...)
- Have few substitutes and are largely complements to the production of private sector goods and services.
- Improvements in the economy's infrastructure generally boost the productivity of private investments.

Infrastructure investment is an important source of productivity growth and should be included as an input in the production function.

[LOS 2.h] Explain how investment in physical capital, human capital, and technological development affects economic growth

4.

Summary of economic growth determinants

Long-term sustainable growth

Real potential GDP growth

Technology improvements

Expansion of supply factors of production (inputs)

Human capital

ICT/non-ICT capital

Labor

Public capital

Natural resources

[LOS 2.i] Compare classical growth theory, neoclassical growth theory, and endogenous growth theory

Growth theories

Classical growth theory

Per capita economic growth is **only temporary** because an exploding population with limited resources brings growth to an end.

Neoclassical growth theory

Long-run per capita growth depends solely on **exogenous technological progress**.

Endogenous growth theory

Explains technological progress **within the model**, rather than taking it as an external factor.

[LOS 2.i] Compare classical growth theory, neoclassical growth theory, and endogenous growth theory

1.

Classical Model

Based on
Malthusian
theory



*Growing
population*



*Limited
resources*

Mechanism

Income per capita > Subsistence income

Technological
progress



Land
expansion
(fixed factor)



Higher
population
growth

*Growing
population*

faces
↓

Diminishing
marginal
return



Produce less
output

*Limited
resources*

Income per capita
returns to subsistence
income

[LOS 2.i] *Compare* classical growth theory, neoclassical growth theory, and endogenous growth theory

1.

Classical Model

Conclusion

Income per capita
>
Subsistence income

Income per capita
=
Subsistence income

New technology results in *larger but not richer* population

Constant living standard overtime

Drawbacks

- In reality, as growth of per capita income $\uparrow$, population growth $\downarrow$
- If technological progress > diminishing marginal returns $\rightarrow$ Possible growth in per capita income

[LOS 2.i] *Compare* classical growth theory, neoclassical growth theory, and endogenous growth theory

2.

Neoclassical Model (Solow)

Objective

Determine the long-run growth rate of output per capita (Y/K) and relate it to:

- Savings/investment rate
- Rate of technological change
- Population growth

Steady-state rate of growth

*Per capita version of
Cobb-Douglas production function*

$$y = Y/L = Ak^\alpha \rightarrow \Delta y/y = \Delta A/A + \alpha \Delta k/k$$

*Rates of change
(capital and output per worker)*

$$k = K/L \rightarrow \Delta k/k = \Delta K/K - \Delta L/L$$

$$y = Y/L \rightarrow \Delta y/y = \Delta Y/Y - \Delta L/L$$

[LOS 2.i] Compare classical growth theory, neoclassical growth theory, and endogenous growth theory

2.

Neoclassical Model (Solow)

Steady-state rate of growth

In the steady state, growth rate of capital per worker equals growth rate of output per worker

$$\Delta k/k = \Delta y/y = \Delta A/A + \alpha \Delta k/k$$

$$\rightarrow \Delta k/k - \alpha \Delta k/k = \Delta A/A$$

$$\rightarrow \Delta k/k (1 - \alpha) = \Delta A/A$$

$$\rightarrow \Delta k/k = \frac{\Delta A/A}{(1 - \alpha)}$$

$$\rightarrow \Delta k/k = \Delta y/y = \frac{\Delta A/A}{(1 - \alpha)}$$

Let $\theta = \Delta A/A$ = growth rate of technology change

$$\rightarrow \Delta y/y = \frac{\theta}{(1 - \alpha)}$$

Equilibrium growth rate of output per capita is a *constant* that depends only on:

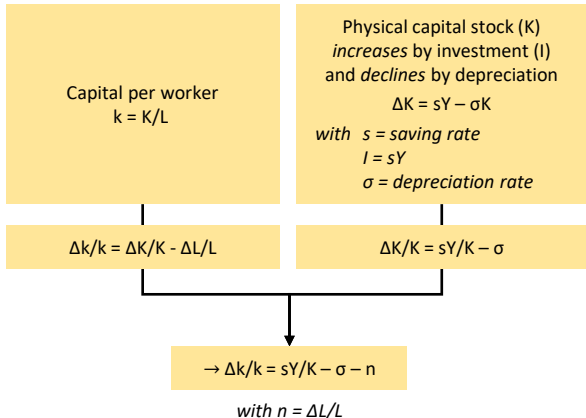
- The growth rate of TFP (θ)
- The elasticity of output with respect to capital (α)

[LOS 2.i] Compare classical growth theory, neoclassical growth theory, and endogenous growth theory

2.

Neoclassical Model (Solow)

Steady-state rate of growth



[LOS 2.i] *Compare* classical growth theory, neoclassical growth theory, and endogenous growth theory

2.

Neoclassical Model (Solow)

Steady-state rate of growth

Growth rate of output per capita: $\Delta y/y = \frac{\theta}{(1-\alpha)}$

Recall from slide 151: $\Delta y/y = \Delta Y/Y - \Delta L/L \rightarrow \Delta Y/Y = \Delta y/y + \Delta L/L$

$\rightarrow$ Growth rate of output $= \Delta Y/Y = \frac{\theta}{(1-\alpha)} + n$ (with $n = \Delta L/L$)

Recall from slide 154: $\Delta k/k = \Delta y/y = sY/K - \sigma - n = \frac{\theta}{(1-\alpha)}$

$$\rightarrow \frac{Y}{K} = \frac{1}{s} \left[\frac{\theta}{(1-\alpha)} + \sigma + n \right] \equiv \Psi$$

Conclusion

1

Output-to-capital ratio is constant (Ψ) and the capital-to-labor ratio (k) and output per worker (y) grow at the same rate $\frac{\theta}{(1-\alpha)}$

2

Marginal product of capital is constant and equal to $\alpha(Y/K)$ and is the real interest rate in the economy

Capital deepening is occurring, but it has *no effect* on the growth rate of the economy or on the marginal product of capital once the *steady state* is reached.

[LOS 2.i] *Compare* classical growth theory, neoclassical growth theory, and endogenous growth theory

2.

Neoclassical Model (Solow)

Steady-state rate of growth

From slide 155: $sY/K - \sigma - n = \frac{\theta}{(1 - \alpha)}$

$$\rightarrow sY/K = \frac{\theta}{(1 - \alpha)} + \sigma + n$$

$$\rightarrow sY = \left[\frac{\theta}{(1 - \alpha)} + \sigma + n \right] \times K$$

Stated in terms of per worker:

$$sy = \left[\frac{\theta}{(1 - \alpha)} + \sigma + n \right] \times k$$

Steady-state equilibrium occurs at the output-to-capital ratio (k) where the savings and actual gross investment per worker generated in the economy (sy) are just sufficient to:

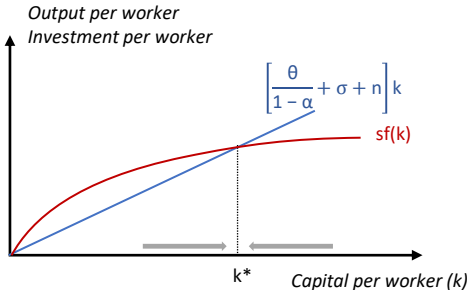
- Provide capital for new workers entering at rate (n)
- Replace plant and equipment wearing out at rate (σ)
- Deepen the physical capital stock at rate $\left(\frac{\theta}{1 - \alpha} \right)$

[LOS 2.i] Compare classical growth theory, neoclassical growth theory, and endogenous growth theory

2.

Neoclassical Model (Solow)

Steady-state rate of growth



The straight line: $sy = \left[\frac{\theta}{1-\alpha} + \sigma + n\right] \times k$

- sy : The amount of investment required
- $\left[\frac{\theta}{1-\alpha} + \sigma + n\right]$: Slope of the line
- k : Capital per worker

The curve line: $sf(k)$

- $sf(k)$: The amount of actual investment per worker
- s : saving rate
- $f(k)$: production function

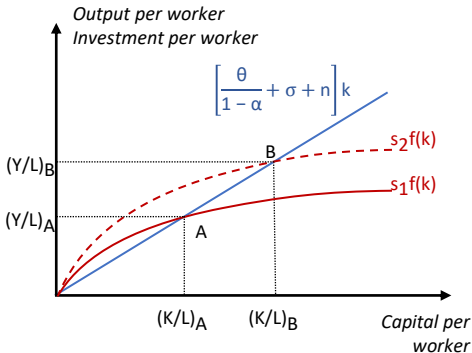
The intersection of the required investment and actual investment lines determines the steady state.

[LOS 2.i] Compare classical growth theory, neoclassical growth theory, and endogenous growth theory

2.

Neoclassical Model (Solow)

Impact of various parameters on the steady state



Saving rate (s)

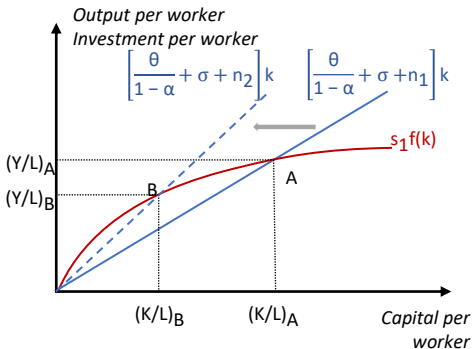
- $sf(k)$ shifts upward
- Capital-to-labor ratio (k) $\uparrow$ and output per worker (y) $\uparrow$ (point B)
- Changes the *level* of output per worker but does not permanently change the *growth rate* of output per worker

[LOS 2.i] Compare classical growth theory, neoclassical growth theory, and endogenous growth theory

2.

Neoclassical Model (Solow)

Impact of various parameters on the steady state



Labor force growth (n)

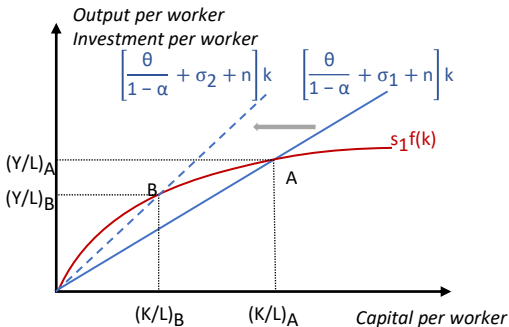
- Slope of the required investment line $\uparrow$
- Equilibrium capital-to-labor ratio $\downarrow$ and output per worker $\downarrow$ (Point B)
- Changes the *level* of output per worker but does not permanently change the *growth rate* of output per worker

[LOS 2.i] Compare classical growth theory, neoclassical growth theory, and endogenous growth theory

2.

Neoclassical Model (Solow)

Impact of various parameters on the steady state



Depreciation rate (σ)

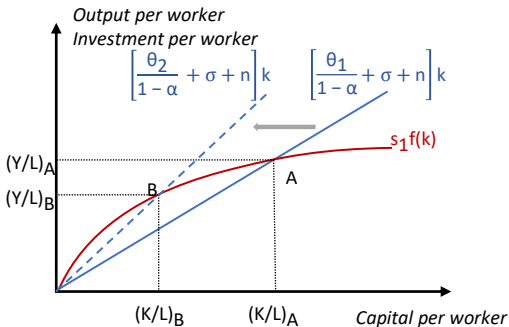
- Slope of the required investment line $\uparrow$
- Equilibrium capital-to-labor ratio $\downarrow$ and output per worker $\downarrow$ (Point B) (*same as labor force growth*)
- Changes the *level* of output per worker but does not permanently change the *growth rate* of output per worker

[LOS 2.i] Compare classical growth theory, neoclassical growth theory, and endogenous growth theory

2.

Neoclassical Model (Solow)

Impact of various parameters on the steady state



Growth in TFP (θ)

- Slope of the required investment line $\uparrow$
- Equilibrium capital-to-labor ratio $\downarrow$ and output per worker $\downarrow$ (Point B) (same as labor force growth)
- Permanently increase the growth rate in output per worker

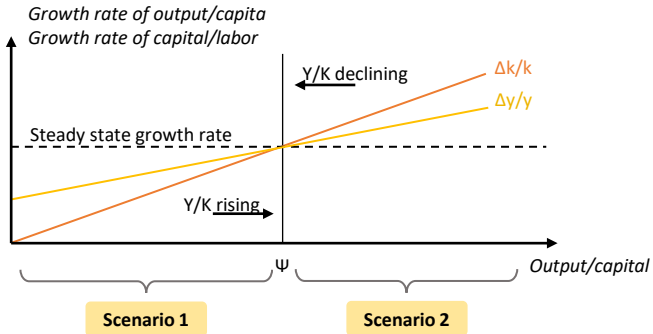
[LOS 2.i] Compare classical growth theory, neoclassical growth theory, and endogenous growth theory

2.

Neoclassical Model (Solow)

Transition to steady-state growth path

During the transition, the economy can experience either faster or slower growth relative to the steady state.



[LOS 2.i] Compare classical growth theory, neoclassical growth theory, and endogenous growth theory

2.

Neoclassical Model (Solow)

Transition to steady-state growth path

Scenario 1

Output-to-capital ratio < Equilibrium level (Ψ)

Growth rates of output per capita <

Steady-state rate $\left(\frac{\theta}{1-\alpha}\right)$

Capital-to-labor ratio < Steady-state rate

$\left(\frac{\theta}{1-\alpha}\right)$

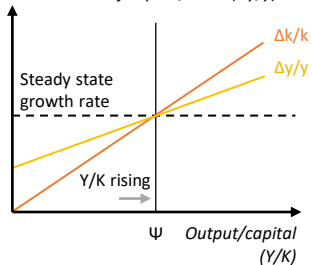
Actual saving/investment < Required investment

Actual investment is *insufficient* to sustain the trend rate of growth in capital-to-labor ratio (k).

Over time, output grows faster than capital, output-to-capital (Y/K) ratio rises, and growth converges to the trend rate.

Growth rate of output/capita ($\Delta k/k$)

Growth rate of capital/labor ($\Delta y/y$)



- $\Delta y/y > \Delta k/k$
- Y/K rises to equilibrium level (Ψ)

[LOS 2.i] Compare classical growth theory, neoclassical growth theory, and endogenous growth theory

2.

Neoclassical Model (Solow)

Transition to steady-state growth path

Scenario 2

Output-to-capital ratio > Equilibrium level (Ψ)

Growth rates of output per capita >

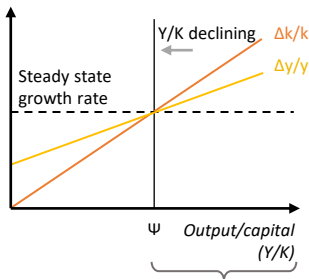
Steady-state rate ($\frac{\theta}{1-\alpha}$)

Capital-to-labor ratio > Steady-state rate

($\frac{\theta}{1-\alpha}$)

Growth rate of output/capita ($\Delta k/k$)

Growth rate of capital/labor ($\Delta y/y$)



Actual saving/investment > Required investment

Above-trend rate of capital deepening leads to above-trend growth in per capita output (y).

Over time, the growth rates of both output per capita (y) and capital-to-labor ratio (k) decline to the steady-state rate.

- $\Delta y/y < \Delta k/k$
- Y/K falls to equilibrium level (Ψ)

[LOS 2.i] *Compare* classical growth theory, neoclassical growth theory, and endogenous growth theory

2.

Neoclassical Model (Solow)

Implications of Neoclassical Model

1. Capital Accumulation

- Affects the *level* of output, *not growth rate* in the long-run.
- A growing economy will move to a point of steady-state growth, *regardless* of initial capital-to-labor ratio or productivity.
- In the steady-state, growth rate of output *does not depend on* capital accumulation or rate of investment. The growth rate of output equals the rate of labor force growth (n) plus the rate of growth in TFP scaled by labor's share of income ($\frac{\theta}{1-\alpha}$).

2. Capital Deepening vs. Technology

- Rapid growth above the steady-state growth rate occurs when countries first begin to accumulate capital, but *growth slows as the process of accumulation continues*.
- Long-term sustainable growth *cannot rely solely* on capital deepening investment.
- The only way to sustain growth in potential GDP per capita is *through technological change or growth in total factor productivity*.

[LOS 2.i] *Compare* classical growth theory, neoclassical growth theory, and endogenous growth theory

2.

Neoclassical Model (Solow)

Implications of Neoclassical Model

3. *Convergence*

- *Growth rates* of developing countries should exceed those of developed countries.
- There should be a convergence of per capita incomes between developed and developing countries over time.

4. *Effect of Savings on Growth*

- During transition, higher saving rate → growth exceeds the steady-state growth rate. After transition period, the economy returns the balanced growth path.
- In steady state, growth does not depend on the percentage of income saved or invested.
- Countries with higher savings rate will have higher level of per capita output, capital-to-labor ratio, and level of labor productivity.

[LOS 2.i] Compare classical growth theory, neoclassical growth theory, and endogenous growth theory

 Knowledge check

Question: Country X has output elasticity of capital of 0.6 and population growth of 2%. If total factor productivity growth is 1%, what is the sustainable growth rate in output according to neoclassical theory?

- A. 2.0%
- B. 2.7%
- C. 4.5%

[LOS 2.i] Compare classical growth theory, neoclassical growth theory, and endogenous growth theory

Knowledge check

Question: Country X has output elasticity of capital of 0.6 and population growth of 2%. If total factor productivity growth is 1%, what is the sustainable growth rate in output according to neoclassical theory?

- A. 2.0%
- B. 2.7%
- C. 4.5%

Answer: C

Recall: According to Neoclassical theory, during steady-state:

- Growth rate of output per capita: $\Delta y/y = \frac{\theta}{(1-\alpha)}$
- Growth rate of output: $\Delta Y/Y = \frac{\theta}{(1-\alpha)} + n$ (with $n = \Delta L/L$)

The question is asking for sustainable growth rate in output

→ Use the second formula, with $\theta = 1\%$, $\alpha = 0.6$, $n = 2\%$:

$$\text{Growth rate of output: } \Delta Y/Y = \frac{\theta}{(1-\alpha)} + n = \frac{1\%}{(1-0.6)} + 2\% = 4.5\%.$$

[LOS 2.i] Compare classical growth theory, neoclassical growth theory, and endogenous growth theory

3.

Endogenous growth Model

Neoclassical growth theory

- Growth in TFP is treated as *residuals*.
- The economy *will eventually reach* steady state.
- *Establish* diminishing marginal return to capital.

Endogenous growth theory

- Technological progress is *explained* rather than treated as exogenous.
- The economy *doesn't necessarily converge* to a steady state rate.
- *No* diminishing marginal return to capital for the economy as a whole.

Endogenous growth theory

- Increasing investment permanently increases the rate of economic growth.
- Possibility of increasing returns to scale.

Definition of "Capital"

Includes human or knowledge capital and R&D (investment in new knowledge improving the production process).

→ These are factors of production like capital and labor, and have to be paid through savings.

[LOS 2.i] *Compare* classical growth theory, neoclassical growth theory, and endogenous growth theory

3.

Endogenous growth Model

Production function

$$y_e = f(k_e) = ck_e$$

Where:

- y_e = Output per worker
- c = Constant marginal product of capital
- k_e = Capital per worker

Constant returns to scale

- Output per worker grows at the same rate as capital per worker.
- Faster or slower capital accumulation translates one for one into faster or slower growth in output per capita.

Recall from slide 153:

$$\Delta k/k = sY/K - \sigma - n$$

In the steady state, growth rate of capital per worker = growth rate of output per worker.

$$\rightarrow \Delta y_e/y_e = \Delta k_e/k_e = \boxed{s - \sigma - n} \rightarrow \text{Constant}$$

→ Higher saving rates imply a permanently higher growth rate.

[LOS 2.i] *Compare* classical growth theory, neoclassical growth theory, and endogenous growth theory

3.

Endogenous growth Model

- There are positive externalities (*societal benefits*) with spending on R&D and knowledge capital.
 - If private sector under-invest in R&D
 - Government intervention (spending on R&D, tax breaks, subsidies) to encourage private production of knowledge capital.
- Constant or increasing returns from investment in knowledge capital → Developed countries can continue to grow as fast as or faster than the developing countries.

The incomes of developed and developing countries do not have to converge.

[LOS 2.j] *Explain and evaluate convergence hypotheses*

Will the difference persist forever, or will the per capita income level of the developing countries converge to those of the developed countries?

Convergence

Types of convergence

Absolute convergence

Conditional convergence

Club convergence

Non-convergence trap

Routes to convergence

Capital deepening

Adoption of new technology

[LOS 2.j] *Explain and evaluate convergence hypotheses*

1. Types of convergence

Convergence means countries with low per capita incomes should grow at a faster rate than countries with high per capita incomes → Converge at some point.

1.1. Absolute convergence

Definition

Developing countries will eventually catch up with the developed countries and match in per capita output, *regardless of their distinct characteristics*.

Note: Neoclassical model does not imply the same *level* of per capita income between countries regardless of underlying characteristics, or it does not imply **absolute convergence**.

[LOS 2.j] *Explain and evaluate convergence hypotheses*

1. Types of convergence

Convergence means countries with low per capita incomes should grow at a faster rate than countries with high per capita incomes → Converge at some point.

1.2. Conditional convergence

Definition

Convergence is conditional on the countries having the same saving rate, population growth rate, and production function.

Note: *If the conditions hold*, the Neoclassical model implies convergence to the same level of per capita output as well as the same steady-state growth rate.

Club convergence

Countries with similar institutions and certain input factors (*in the club*) will eventually converge.

Non-convergence trap

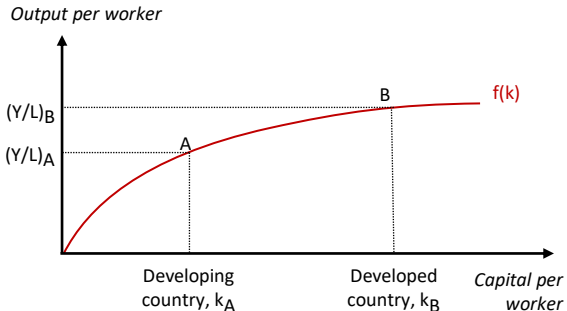
Countries may fall into *non-convergence trap* if they do not implement necessary institutional reforms.

Investing in countries with *lower* per capita incomes that are *members of the club* should provide a higher rate of return in the long run.

[LOS 2.j] *Explain and evaluate* convergence hypotheses

2. Routes to convergence

2.1. Capital accumulation and capital deepening



Increases in capital significantly **boost** labor productivity.

Increases in capital have **almost no impact** on productivity.

[LOS 2.j] *Explain and evaluate* convergence hypotheses

2.

Routes to convergence

2.2. Imitation or adoption of technology



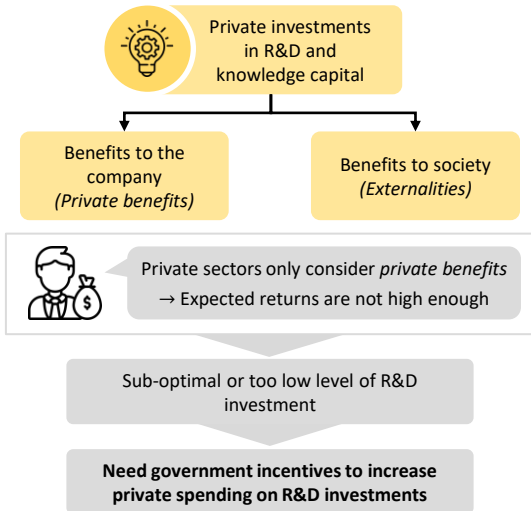
Technology transfers: Developing countries import technology from the advanced countries

Join the convergence club

Narrow the income gap

[LOS 2.k] Describe the economic rationale for governments to provide incentives to private investment in technology and knowledge

According to endogenous growth theory



[LOS 2.1] Describe the expected impact of removing trade barriers on capital investment and profits, employment and wages, and growth in the economies involved

Neoclassical model

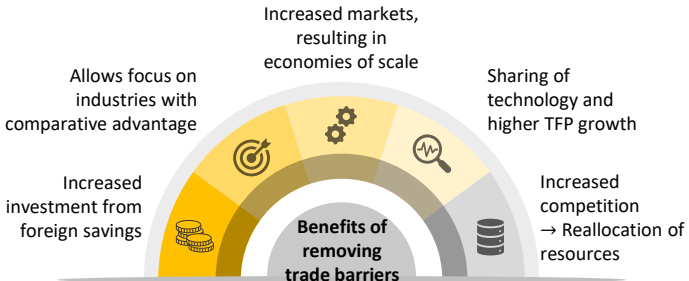
- Closed economy: Domestic investment = Domestic savings
→ Limited investment
 - Open economy: Investment is not constrained by domestic savings
- Convergence should occur more quickly if economies are open and there is free trade and international borrowing and lending.

Endogenous growth model

- More open trade policy will permanently raise the rate of economic growth since open markets foster increased innovation.
- International trade increases global output through the following:
- **Selection effect:** Increased competition from foreign companies forces *less efficient* companies to *exit* and *more efficient* ones to *innovates and raises the efficiency*.
 - **Scale effect:** Producers *exploit economies of scale* by selling to a larger market.
 - **Backwardness effect:** Less advanced countries or sectors of an economy *catching up* with the more advanced countries or sectors *through knowledge spillovers*.

[LOS 2.1] Describe the expected impact of removing trade barriers on capital investment and profits, employment and wages, and growth in the economies involved

Benefits of removing trade barriers





MODULE 3: ECONOMICS OF REGULATION

LEARNING OUTCOMES

- 3.a.** Describe the economic rationale for regulatory intervention
- 3.b.** Explain the purposes of regulating commerce and financial markets
- 3.c.** Describe anticompetitive behaviors targeted by antitrust laws globally and evaluate the antitrust risk associated with a given business strategy
- 3.d.** Describe classifications of regulations and regulators
- 3.e.** Describe the use of self-regulation in financial markets
- 3.f.** Describe regulatory interdependencies and their effects
- 3.g.** Describe tools of regulatory intervention in markets
- 3.h.** Describe benefits and costs of regulation
- 3.i.** Describe the considerations when evaluating the effects of regulation on an industry

[LOS 3.a] *Describe* the economic rationale for regulatory intervention

Question: *Is regulatory intervention necessary in an economy?*



Market failures

Can be corrected by market solutions



But market solutions are not adequate for all market situations

Regulations are necessary

[LOS 3.a] *Describe* the economic rationale for regulatory intervention

1. Informational friction

Definition

Asymmetrical information: Lack of access to information and inadequate information → Adverse selection and moral hazard.
E.g.: If a homeowner wanted to sell their house, they would have more information about the house than the buyer.

Regulation

Ensure that no participant is disadvantaged.

2. Externalities

Definition

Externalities (Spillover effects): The costs and benefits from production and consumption activities experienced by a third party.

- Positive externalities → Spillover benefits
- Negative externalities → Spillover costs.

E.g.: Home improvements → Neighbors may benefit from the increase in home values in the neighborhood.

Regulation

Mitigate the consequences from the negative externalities for the public.

[LOS 3.a] *Describe* the economic rationale for regulatory intervention

3. Weak competition

Definition

Consumers owe to high prices, less choice, and lack of innovation, mainly due to dominant firm has significant market power.
E.g.: Using long-term or exclusive contracts to stop customers from changing suppliers

Regulation

Antitrust regulations.

4. Social objectives

Definition

Achieved via provision of public goods whose consumption by one individual does not reduce the availability of the good for others.
E.g.: Telecommunication firms provide service to remote customers

Regulation

Placing regulatory obligation on firms to ensure customers are served regardless of the additional costs.

[LOS 3.b] Explain the purposes of regulating commerce and financial markets

1.

Regulation of Commerce

Government policy can be important for promoting commerce locally, nationally, regionally, and globally. These policies are created to regulate issues such as:



Trade agreement



Basic features of the business environment



Labor markets



Immigration issues



Recognition and protection of intellectual property



Setting common technical standards



Fundamental safety



Operation of commerce

[LOS 3.b] Explain the purposes of regulating commerce and financial markets

2.

Regulation of Financial market

The regulation of securities markets and financial institutions is essential because of the consequences to society of failures in the financial system.

These regulations address problems in the financial market such as:



Enhances confidence in the financial system



Protection of consumers and investors



Safety and soundness of financial institutions



Enhancing capital formation



Securities registration requirement



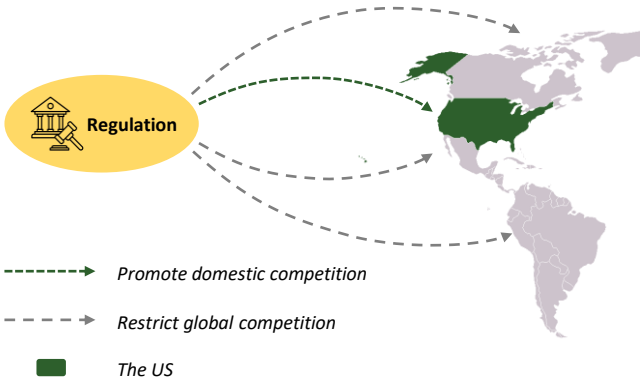
Disclosure requirement

[LOS 3.c] Describe anticompetitive behaviors targeted by antitrust laws globally and evaluate the antitrust risk associated with a given business strategy

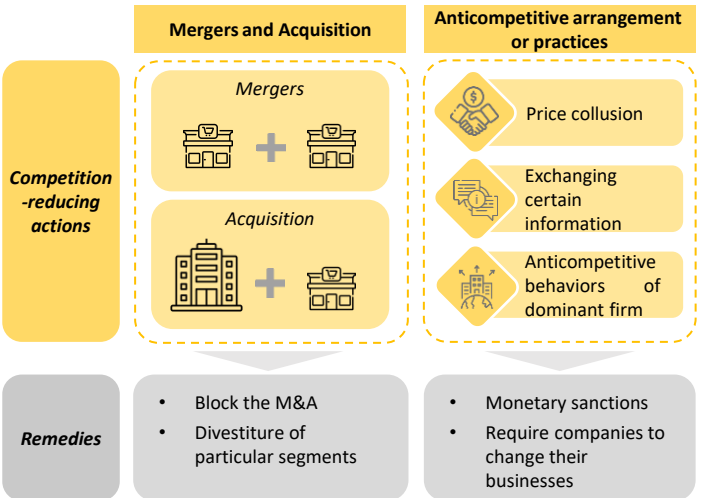
1.

Regulatory goal of the government

In a global context, an implicit regulatory goal of government may be to restrict competition from other countries.



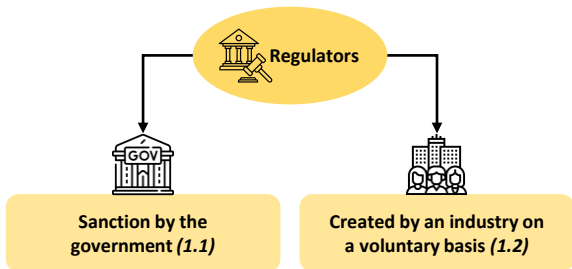
[LOS 3.c] Describe anticompetitive behaviors targeted by antitrust laws globally and evaluate the antitrust risk associated with a given business strategy

2.
Competition and Antitrust laws


**[LOS 3.d,e] *Describe* classifications of regulations and regulators;
Describe uses of self-regulation in financial markets**

1.

Classification of Regulations and Regulators



**[LOS 3.d,e] *Describe* classifications of regulations and regulators;
Describe uses of self-regulation in financial markets**

1.

Classification of Regulations and Regulators

1.1. Government sanctioned regulators

Regulators

Legislative bodies

Government agencies

Independent regulators*

Courts

Regulations

Statutes

Administrative regulations,
Administrative laws

Judicial law

*Enacted at
broad level*
*Applied in
detail*
Interpreted

Regulatory bodies *have legal authority* to enact and enforce regulations.

* *Independent regulators:*

- *Not relying* on government funding.
- Immune from political influence and pressure → *Autonomy* in decision making.
- Authority comes from the *recognition given* by a government agency.

**[LOS 3.d,e] Describe classifications of regulations and regulators;
Describe uses of self-regulation in financial markets**

1.

Classification of Regulations and Regulators

1.2. Industry created (Private) regulators



**Self-regulatory
bodies**

- Private organizations that both represent and regulate their members.
- Isolated from political pressure.
- Derive authority from members who agree to comply with the rules and standards → Subject to pressure from members.

+ *Recognition and authority
from the government*



**Self-regulating
organizations**

- Represent and regulate their members.
- Has the force of law.
- Funded independently.

E.g.: US SEC (government agency) allocates some regulatory responsibilities to FINRA (an SRO).

**[LOS 3.d,e] Describe classifications of regulations and regulators;
Describe uses of self-regulation in financial markets**

2.

Comparing regulatory bodies

	Government-backed bodies*	Independent regulators	Self-regulatory bodies	Self-regulating organizations
Funding	Funded by the government	Funded independently		
Authority	Given by a statute, a government department or agency. → <i>Has the force of law.</i>	Given by a statute, a government department or agency. → <i>Has the force of law.</i>	Derive authority from members who agree to comply with the rules and standards. → <i>Do not have the force of law.</i>	Similar to self-regulatory bodies. Recognition and authority with enforcement power given by a government body or agency → <i>Has the force of law.</i>

*for comparison purpose, "Government-backed bodies" do not include independent regulators.

**[LOS 3.d,e] *Describe* classifications of regulations and regulators;
Describe uses of self-regulation in financial markets**

 Knowledge check

Example: The media devotes considerable coverage to a regulatory body that has been given autonomy by the government and is empowered by statute. The regulatory body has recently raised the fees charged to the companies it regulates. The regulatory body in question is most likely to be a(n):

- A. Self-regulatory organization.
- B. Government agency.
- C. Independent regulator.

**[LOS 3.d,e] Describe classifications of regulations and regulators;
Describe uses of self-regulation in financial markets**

Knowledge check

Answer: C

Recall: Comparison of regulators

	Government agency	Independent regulators	Self-regulatory organization
Funding	Funded by the government	Funded independently	Funded independently
Authority	Given by the government → <i>Has the force of law.</i>	Given by the government → <i>Has the force of law.</i>	Similar to self-regulatory bodies, but has the authority given by the government. → <i>Has the force of law.</i>

The characteristics of the regulatory given in the text:

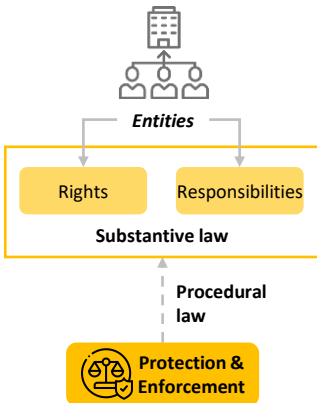
- Autonomy given by the government, empowered by statute → Authority given by the government
- Charge fees to companies it regulates → Funded by fees collected or funded independently from the government.

→ Matches with **Independent regulators**.

**[LOS 3.d,e] Describe classifications of regulations and regulators;
Describe uses of self-regulation in financial markets**

3.

Classification of laws and regulation



- **Substantive law:** Focuses on the *rights and responsibilities* of entities and relationships among entities

→ What to regulate?

- **Procedural law:** Focuses on the *protection and enforcement* of the substantive laws

→ How to regulate?

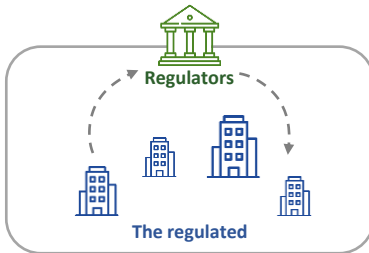
Regulators typically have responsibility for both substantive and procedural aspects of their regulations.

[LOS 3.f] Describe regulatory interdependencies and their effects

1. Regulatory capture theory

Regulatory capture theory: Regulations often arise to enhance and work to the benefit of the interests of the regulated.

- Some individual regulators may *have worked* in the industry
- Some individual regulators may *aspire* to be in the industry



- Regulated entities may possess *considerable expertise and knowledge*

These interactions may *reinforce* regulatory capture.

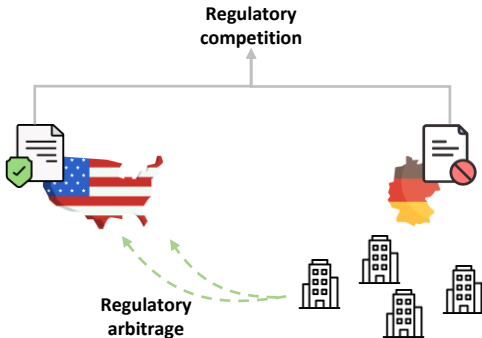
[LOS 3.f] Describe regulatory interdependencies and their effects

2.

Regulatory competition and Regulatory arbitrage

Regulatory competition: Regulators may compete to provide a regulatory environment designed to attract certain entities.

Regulatory arbitrage: Companies pick the country with the easiest/least strict regulations to do business in.



Tips: Regulatory competition – From regulators' perspective

Regulatory arbitrage – From companies' perspective

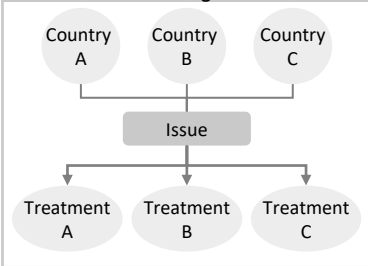
[LOS 3.f] Describe regulatory interdependencies and their effects

3.

Regulatory interdependencies

Diverse regulators among countries

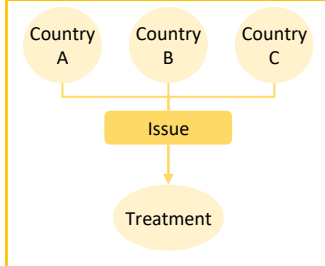
Varying perspective of domestic regulators



Regulatory competition

- “Race to the bottom”;
- The affected stances of national and regional regulators;
- Regulatory arbitrage

Regulatory cooperation/coordination



Governments recognize the necessity for global **regulatory cooperation and coordination** on some issues.

E.g.: Basel Committee on Banking Supervision – standard setter for bank supervision.

[LOS 3.f] Describe regulatory interdependencies and their effects

3.

Regulatory interdependencies

Diverse regulators within 1 country



Securities regulators

- Protect investors
- Ensure markets are fair, efficient and transparent
- Reduce systemic risk

Advocate for the release of information relevant to decision making to protect investors



Bank supervisors

- Safety and soundness of financial institutions
- Promote financial stability, reduce system-wide risks, protect customers

Reluctant to release the results of stress tests to preserve the confidence of the market



Inconsistence

Conclusion: Regulations by different regulators (*on a global or domestic scale*) with seemingly similar objectives can lead to very different regulatory outcomes.

[LOS 3.f] *Describe* regulatory interdependencies and their effects **Knowledge check**

Example: A country's securities regulator is looking to attract a higher number of smaller companies to its capital markets. It proposes to ease hurdles that companies face when preparing to list shares on the country's stock exchange. The proposals include the lowering of the frequency of financial reporting, reducing the extent of disclosures required, and reducing the minimum size of company that can be accepted on the market.

This is an example of:

- A. Regulatory competition.
- B. Regulatory coordination.
- C. Regulatory capture.

[LOS 3.f] *Describe regulatory interdependencies and their effects*



Knowledge check

Answer: A

Recall:

- *Regulatory competition: Regulators may compete to provide a regulatory environment designed to attract certain entities.*
- *Regulatory coordination: cooperation at a global level to achieve a cohesive regulatory framework to avoid regulatory arbitrage.*
- *Regulatory capture: Regulations often arise to enhance and work to the benefit of the interests of the regulated.*

Regulator's purpose: Attract a higher number of smaller companies to its market.

Regulator's action:

- Lowering the frequency of financial reporting
 - Reducing the extent of disclosures required
 - Reducing the minimum size of company that can be accepted
- Providing a less strict environment to attract more companies
- Regulatory competition

[LOS 3.g] *Describe* tools of regulatory intervention in markets

1. Tools of regulatory intervention

Price mechanisms

Taxes

Create appropriate marginal incentives and an efficient allocation of resources.

Subsidies

Mandates & Restrictions

Regulatory mandates

What the regulated must do
E.g.: Capital requirements for banks, ...

Restrictions on behaviors

What the regulated must not do
E.g.: Insider trading, short selling, ...

Provision of public goods

Providing public goods.
E.g.: National defense, public infrastructures, ...

Public financing of private projects

Funding specified activities that the government desires to encourage.
E.g.: Student loans or small-business loans, ...

Depend on political priorities and objectives of the government

Sometimes, a combination of regulatory tools may work better in conjunction.

[LOS 3.g] *Describe* tools of regulatory intervention in markets

2.

Other aspects of regulatory tools

Negative externalities

Systemic risk

The risk of failure in financial system.

Financial contagion

Financial shocks spread from their place of origin to other regions.

Challenges in regulation designing

- Regulations designed with prior crisis may not prevent future crisis.
- Regulatory responses may have unintended consequence of mitigating one source of risk while increasing the other.

Effectiveness of regulation

Depends on the ability to enforce the regulations.

[LOS 3.g] *Describe* tools of regulatory intervention in markets

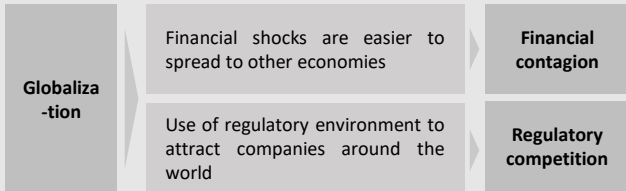
Knowledge check

Example: Globalization of capital markets is most likely to result in increased concerns about:

- A. Financial contagion.
- B. Regulatory competition.
- C. Both contagion and regulatory competition.

Answer: C

Recall: Globalization is the process of interaction and integration among people, companies, and government worldwide (CFA Curriculum level I 2023).



[LOS 3.h] *Describe* benefits and costs of regulation

From regulated entities' perspective:

Regulatory burden: The costs of regulation for the regulated entity, sometimes viewed as the *private costs* of regulation or *government burden*.

Private cost
of regulation

—

Private
benefits
of regulation

=

Net regulatory
burden

From regulators' perspective:

Costs of regulation

Direct costs

+

Indirect costs

+

Unanticipated
costs

Unintended costs

Implementation
costs of
regulation

Costs associated with changing
economic decisions and behavior,
and varying market allocation

- Regulatory costs and benefits *are difficult to assess* on a prospective basis relative to retrospective basis.
- An *after-the-fact* (retrospective) analysis allows a comparison of the items of interest before and after the regulation occurs.
- A *pilot study* can be done to analyze the potential impact of a proposed regulation.

[LOS 3.i] *Describe* the considerations when evaluating the effects of regulation on an industry

From regulated entities' perspective:

Evaluating effects of regulation

1. Assessment of the likelihood of regulatory change

- Understanding the regulator's intentions.
- Cost-benefit analysis framework.
- The extent of engagement with the regulated companies.
- Public and political pressure.

2. Assessment of the impact of regulatory change on a sector

Impact on revenues

- Limits on prices, tariffs, rents, fees to protect consumers;
- Prohibition of products or services; ...

Impact on costs

Compliance with regulations results in additional costs
→ *Compliance costs*.

Business risk

Uncertainty is hard to quantify
→ *Assign probabilities* to events or reflect the risk in the *higher discount rate* to value the firm.

[LOS 3.i] *Describe* the considerations when evaluating the effects of regulation on an industry



Knowledge check

Example: Jessica Wong, CFA, is an equity analyst responsible for the materials and industrial sectors in Europe. The regulatory authorities are preparing new rules on transportation, further limiting the age and exhaust emissions of the trucks used by industrial companies.

What is likely to be of greatest concern to the analyst when evaluating the impact of the new rules on companies? This is an example of:

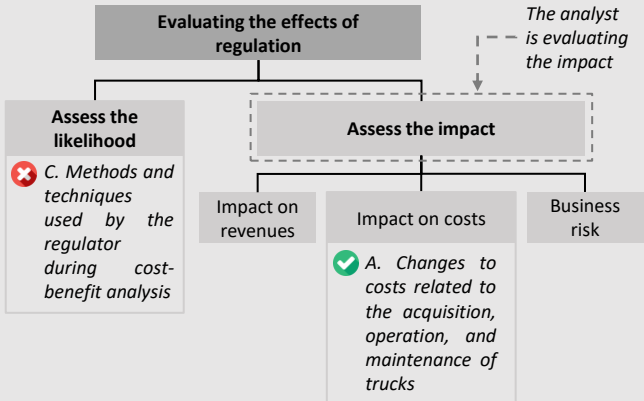
- A. Changes to costs related to the acquisition, operation, and maintenance of trucks.
- B. Positive impact of reduced pollution on public health and subsequent health care cost savings in the wider society.
- C. The methods and techniques used by the regulator during its cost-benefit analysis.

[LOS 3.i] Describe the considerations when evaluating the effects of regulation on an industry

Knowledge check

Answer: A

Recall: Evaluating the effects of regulation includes:



✗ B. Positive impact of reduced pollution on public health and subsequent health care cost savings in the wider society → Matter of concern for the regulator authority, the question is asking in the analyst's perspective.

SUMMARY

ECONOMICS LEVEL II

Currency exchange rates (module 1)

- Calculate exchange rates
- Spot and forward rates
→ International parities
- Carry trade
- Influencing factors
- Government intervention
- Warning signs

Economic growth (module 2)

- Developed vs. Developing countries
- Stock market vs. the Economy
- Forecasting potential GDP
- Influencing factors
- Growth theories
- Convergence hypotheses
- Government intervention

Economics of Regulation (module 3)

- Rationale for regulation
- Purpose, anticompetitive behavior
- Classification of regulations and regulators
- Interdependencies
- Tools
- Cost-benefit analysis
- Effects on an industry